Optimal Information Structures for Strategic Communication*

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Abstract

This paper considers optimal information structures in restricting the information available to the Sender in a cheap talk game. Monotone partitional information structures might not obtain optimality, and some signals that the Sender receives are required to be partially garbled together in order to relax the Sender's incentive constraints. We provide a partial characterization of the optimal structure and show that it belongs to a class of structures that are more general than monotone partitional, but that only exhibit certain local non-monotonicity of the information structure. Our analysis builds on the graphical characterization of feasible information structures in Gentzkow and Kamenica (2016) and shows that this methodology can be used in complex information design problems with large state and action spaces. As an extension we use the same approach to solve problems with capacity constraints on communication.

Keywords: cheap talk, information design, optimal information restriction
JEL Codes: C72; D82; D83

1 Introduction

There are many economically important contexts in which an uninformed decision maker needs to seek advice from an informed but biased expert before making a decision, such as when a policy-maker consults with an advisor with specialized knowledge or when corporate headquarters gather information from a particular organizational unit. Starting with the seminal work of Crawford and Sobel (1982), there is a large literature investigating such interactions focusing on the case when

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communication between the parties takes the form of cheap talk: an informed Sender sending a cheap talk message to a Receiver who subsequently chooses an action.[1] Most of this literature assumes that the information available to the Sender is exogenously given. However, there are many settings in which the Receiver can restrict the information that the Sender can learn before the communication takes place. For example, organizational headquarters can put restrictions on what processes the head of an organizational unit can monitor and keep records on. Or a regulatory agency collecting information on demand conditions from firms at a market can restrict the type of data the firms are allowed to collect on their consumers. More recently Fischer and Stocken (2001) and Ivanov (2010b) showed that such reduction of the quality of the information available to the Sender can be beneficial for the Receiver (and indeed to both parties from an ex ante point of view). The main intuition is that coarser information can ease the incentive constraints of the Sender and hence can facilitate credible communication of more information.[2]

The main focus of the current paper is characterizing the optimal way of restricting the information available to the Sender.[3] Ivanov (2010b) characterizes the optimal information structure in the uniform-quadratic specification of the Crawford and Sobel model subject to the assumption that the optimal information structure is monotone partitional, but leaves the question open whether restricting attention to such information structures results in sub-optimality. We show that indeed it is suboptimal, even in the uniform-quadratic specification of the cheap talk model. Our leading example features a range of bias parameters of the Sender for which the optimal partitional structure involves partitioning the state space (the unit interval in $\mathbb{R}$) to three equal size cells and the Sender only learning which cell the state falls in, or equivalently only learning the conditional expectation of the state given the cell where the state is. For biases in the range we consider in this example, there is no partitional structure with more than three cells for which it is incentive compatible for the Sender to truthfully reveal the observed cell for all of the possible cell realizations. Inevitably, when partitioning the state space into four or more cells, conditional expectations of at least two neighboring cells would be close enough to each other such that the biased Sender would have an incentive to claim that the state is from the neighboring cell, not from the one that the Sender observed. On the other hand, it is possible to construct non-partitional information structures with four signal realizations, each signal associated with different conditional expectations

[1] See also Green and Stokey (2007) the first version of which was written around the same time.

[2] Related to this, Austen-Smith (1994) demonstrates that making information acquisition costly for the Receiver can improve the efficiency of communication, and Ivanov (2010a) and Ambrus et al. (2013) show that communication via strategic Mediators, where the information of a Mediator is endogenously coarse relative to the Sender, can have similar effects.

[3] In our context ex ante the Sender and the Receiver have aligned interests over information structures (both would like to minimize the conditional variance of the Receiver's action choice relative to the true state), hence the optimal information structures are the same for both players. Therefore, our results are valid no matter whether the Sender or the Receiver chooses the information structure.

of the state, such that the Sender can truthfully reveal his information to the Receiver without violating incentive compatibility constraints. These structures resemble a partitional structure with four information cells, but with some garbling of the information from the two middle cells: with some small probability the signal of the Sender being the second highest one even when the true state is from the third highest cell, and vice versa. Relative to a partitional structure restricted on the same states, this garbling pulls the conditional expectations associated with the two signals closer to each other, away from the conditional expectations associated with the extreme signals. This in turn makes it possible to add two other information cells at the extremes without violating incentive compatibility. The trade-off associated with this is that while a higher number of signals is credibly conveyed to the Receiver, the quality of some of the signals is reduced by the garbling in the middle, but for a range of biases the garbled information structure with more signals yields higher payoffs.

For an example how such garbling of the Sender's information can be welfare improving, consider a search committee in an academic department in a setting where a job candidate can only be evaluated by the committee member who is from the same field as the candidate. Assume that candidates can be lumped into four categories: Unacceptable, Mediocre, Good, and Excellent, and ideally the committee would like to learn from the expert committee member which category the candidate falls into. However, the committee member is biased towards her own field and has an incentive to claim that a candidate is Excellent even when she knows that the candidate is only Good. In this case, instead of providing information to the committee member that lets her accurately identify a candidate's category, it might be better to provide information that garbles the Mediocre and Good categories to some extent: in particular one which implies that with positive probability some Mediocre candidates seem Good to the committee member. This makes the committee member less positive about candidates that seem Good to her, and with enough garbling the committee member does not have incentives to pretend that she observed an Excellent signal instead of a merely Good one (assuming that her bias is not too large). The upside is that the committee can learn the true information of available to the expert member, while the downside is that the expert's information is of lower quality by identifying some Mediocre candidates as Good and vice versa. But if the required garbling is not too much, it improves welfare relative to an alternative information structure that allocates candidates only into three ranked categories and accurately indicates to the the expert member to which category the candidate belongs.

Our main result is a partial characterization of the optimal information structure for quadratic preferences and general state distributions.[4] Just like in Ivanov (2010b), we can restrict attention to information structures in which the signals are just conditional expectations of the state that are

[4]We are motivated by examples in which the information designer is the Receiver, but with quadratic preferences the optimal information structures are the same for the Sender and the Receiver: they are the ones that minimize the conditional variance of the state around implemented actions.

far away from each other such that the Sender is willing to truthfully convey them to the Receiver (implying a finite upper bound on the number of signals, determined by the magnitude of the Sender's bias). We show that the optimal information structure always takes the following form. The state space is partitioned into a finite number of interval regions, and there are two types of regions. The first one is standard information partition cells, in which the same signal is generated from every state of the cell, and this signal is not generated at any state from outside the cell. The second one is interval regions in which two different signals are generated, in a non-monotone fashion, that is, the region cannot be divided into two subintervals such that each signal is only generated in one subinterval. Moreover, both of these signals are only generated within the region. Therefore, the type of garbling in optimal structures is limited to garbling information between two neighboring signals. It cannot involve more complicated non-monotone garbling three or more signals. A special case of this class of information structures is a standard partitional structure in which each region is an information cell generating only one signal. We also show that the lowest and highest region in an optimal information structure always has to be an information cell generating only one signal, and the Sender's incentive compatibility constraints must bind around a garbling region.

The partial characterization result can also be used to characterize exactly the optimal information structure for concrete specifications of the cheap talk game. We do this for the uniform-quadratic specification of the game for biases that are not too small. We show that regions of parameter values alternate for which the optimal information structure is monotone partitional and for which the optimal information structure requires regions with garbled information.

Our analysis builds on the graphical characterization of all feasible information structures in terms of integrated cumulative distribution functions of conditional expectations (posterior means) in Gentzkow and Kamenica (2016), which in turn combines insights from Blackwell (1953) and Rothschild and Stiglitz (1970). In our setting with a finite number of signals (implied by the incentive compatibility conditions), these cumulative distribution functions are piecewise linear, making it easier to work with them. The analysis shows that the graphical characterization of Gentzkow and Kamenica can help solving information design problems with large state and action spaces in more general settings than those they considered — it is not necessary to restrict the Sender's preferences over the Receiver's actions to be state-independent. Also, certain graphical features of integrated cumulative distribution functions can be associated with the features of information structures, allowing the graphical method to be utilized when the number of posterior means is larger than what Gentzkow and Kamenica originally shown.

We further demonstrate the usefulness of working with integrated cumulative distribution functions in two extensions of the model. The first one allows for a class of state-dependent bias function for the Receiver (affine bias), while the second one allows for a finite number of mass points in the

prior distribution, a special case of which is finite discrete state spaces. In case of such mass points the receiver has an incentive to anchor the signal structure around them, as revealing the corresponding states increases payoffs discretely, but this can also tighten the IC constraints for neighboring signals, leading to a trade-off. We also show how the integrated cumulative distribution function approach can be used in solving information design problems in which instead of incentive compatibility constraints on the side of the Receiver, participants face exogenous technological constraints, either in the form of an upper bound on the number of messages that can be used in communications, or an upper bound on the entropy of the information structure (motivated by having a budget constraint on communication in a setting where more informative structures are more costly).

Our paper, although independently written, is closely related to Kreutzkamp and Lou (2025). Subject to technical differences, they analyze the same model, and obtain a set of results largely overlapping with ours. The main difference between the papers is that they use completely different techniques, from the literature on extreme points and majorization,[5] instead of working with the graphical representation of feasible information structures as the current paper. As mentioned earlier, what we regard as one of the main contributions of the current paper is showing how the latter techniques can be used in problems with rich state spaces and state-dependent preferences, which we further demonstrate in the extensions.

A paper using similar techniques as ours, in a different context that combines information design and voluntary disclosure, is Shishkin (2024). In particular, it investigates a setting in which a Sender is trying to persuade a Receiver to accept a project, and the signal realization produces hard evidence with some probability. The Sender then can decide whether to disclose this hard evidence. Our setting differs from this setting in many aspects, one being that the Receiver has a much richer action space, and we demonstrate that the techniques introduced in Gentzkow and Kamenica (2016) can be used in such settings too. More generally, our work builds on insights from the literature on Bayesian persuasion starting with Rayo and Segal (2010) and Kamenica and Gentzkow (2011). In the latter literature Kolotilin (2018), Dworczak and Martini (2019), Ivanov (2021), and Mensch (2021) identify sufficient conditions for the optimality of monotone partitional information structures.

Another paper investigating optimal information design in a cheap talk setting is Krähmer (2021). The main difference between the approaches is that the latter paper allows the principal to design information structures that give him information about the realization of a random variable that the agent does not learn. Equivalently, he can randomize between information signals for the expert in a way that the expert does not know the scope of information she can possibly learn, but the principal does. We instead focus on situations in which the expert has substantially superior

[5] See Kleiner et al. (2021) for an introduction to these techniques.

knowledge and expertise relative to the principal, and assume that it is not feasible for the principal to randomize the information structure in a way that provides him extra information relative to the expert. That is, the outcome of any randomization in the mechanism is either not observed directly by either party, or if it is observed by the principal, then it is also observed by the expert: the latter, using her superior knowledge of the environment, fully recognizes what information structure got randomly selected. In Krähmer (2021), the principal can use his superior information to cross-check reports from the agent, which allows him to achieve the first best, full revelation of information.[6] In contrast, in our model information design improves over communication with unconstrained information on the side of the expert, but the principal's payoff is bounded away from the first best.

For an earlier paper that analyzes how the amount of information transmitted in cheap talk depends on the Sender's information in a finite state space context, see Fischer and Stocken (2001). For a more recent paper investigating information design in a finite state space context see Lyu and Suen (2024).

A paper obtaining some similar results to ours in a different setting is Arieli et al. (2023). They show that in a class of persuasion problems (with no cheap talk component hence no constraints imposed by sequential rationality) the optimal information structure has similar features as in our model: the state space is partitioned into intervals such that at most two signals are associated with each interval. They call this feature bi-pooling. Again they work with a different methodology to obtain this result.

Lastly, for a setting in which the optimal information structure is partitional, see Lou (2023).

2 The Model

There are two players, a Sender and a Receiver. The state of the world is stochastic, distributed according to a commonly known prior distribution F with support $\Theta = [0, 1]$. Suppose that F has a continuous density function f bounded away from 0. Let μ be the prior mean.

The timeline of the game is as follows. At the beginning of the game, the Receiver first chooses an information structure $\pi : \theta \to \Delta(S)$, where S is the set of possible signal realizations. The true state of the world θ then realizes according to F, and a signal realization is sent to the Sender according to $\pi(\cdot \mid \theta)$. The Sender then sends a message $m \in M$ to the Receiver, where M is the set of possible messages. After receiving m, the Receiver then chooses an action $y \in \mathbb{R}$. That is, the information control game is a classic cheap talk game, with an additional information design layer at the beginning. Compared with the classic cheap talk model, this additional information design layer now grants some power to the Receiver to control the information.

[6] See also Watson (1996) for a similar construction.

Given the state θ and the action a, the Sender and the Receiver's utilities are quadratic,

$$U^S(y, \theta, b) = -[y - (\theta + b)]^2 \quad \text{and} \quad U^R(y, \theta) = -(y - \theta)^2,$$

where b represents the Sender's state-independent bias in the perspective of the Receiver. Without loss of generality, we assume that $b > 0$. As Lou (2023) pointed out, and as we will show in detail later, such preferences are equivalent to preferences that are linear in states. For example, we can replace the Receiver's utility as

$$V^R(y, \theta) = 2\theta y - y^2.$$

This equivalence offers special convenience when considering the information design problem (Gentzkow and Kamenica, 2016). The key implication is that we can consider the distribution of posterior means in the information design phase, instead of the complete distribution of posterior beliefs. This reduces the dimension of posteriors to be considered and in turn simplifies the problem.

The equilibrium concept used is perfect Bayesian equilibrium (PBE). In general, a Sender's strategy is a map from the set of signal realizations S to the set of probability distributions $\Delta(M)$. A Receiver's strategy consists of two parts, an information structure π, and a map from the set of messages M to the set of probability distributions $\Delta(Y)$ on the set of possible actions Y. But we can vastly reduce the generality of this problem due to the quadratic preferences. First, there is no need to consider mixed actions. For any posterior belief induced by a signal realization, the Receiver has a unique optimal action given the quadratic preference. Second, as Ivanov (2010b) shows, a revelation-principle type of argument establishes that we can without loss of generality consider truth-telling equilibria in which the Sender truthfully reveals her posterior mean after receiving a signal realization. Given that the Sender reports the truth, the Receiver should always choose the reported posterior mean as the optimal action. For this reason, it is also equivalent to consider the Sender as the information designer in this game. One can consider the Sender recommending actions to the Receiver by truthfully reporting the posterior means, and the Receiver is willing to follow the recommendations.

In order for the Sender to truthfully report her posterior mean, the information structure must be finite, meaning that it can generate at most finitely many posterior means. To see this, suppose that x and x' are two of the posterior means generated under π. In order for the Sender who gets the posterior mean x not to report x', it must be the case that

$$|x - x'| \geqslant 2b.$$

Since x and x' are arbitrarily chosen, this means there must be a distance of at least $2b$ between any two posterior means, which further implies that one can only fit finitely many posterior means in $[0, 1]$. Therefore, it suffices to consider information structures that induce finitely many posterior means. For a given bias b, let $N_{\max}(b) = \lceil 1/2b \rceil$ be the maximum number of posterior means that

can be fitted within $[0,1]$. For an information structure π that induces N posterior means, let the posterior means be ordered

$$0 < x_1 < \cdots < x_N < 1.$$

Although the Receiver can use as many signal realizations as possible to induce these posterior means, it suffices for the Receiver to use N signal realizations. In particular, the Receiver can combine the signal realizations that induce the same posterior means into a single new signal realization. Let s_i be the signal realization that induces x_i. When looking for the optimal information structure, we can restrict attention to structures satisfying the incentive compatibility constraints in terms of posterior means

$$x_i - x_{i-1} \geqslant 2b, \quad i = 2, \cdots, N, \tag{IC}$$

that maximizes the ex-ante expected utility of the Receiver

$$\mathbb{E}U^R(\pi) = \sum_{i=1}^{N} \int_{\Theta} -(x_i - \theta)^2 f(\theta)\pi(s_i \mid \theta)\mathrm{d}\theta. \tag{1}$$

Notice that truthful reporting by the Sender, and the optimal action choice of the Receiver given the truthful report, are already incorporated in (1).

Before we start the formal analysis, note that our goal is to find a Receiver-optimal information structure in terms of ex-ante payoffs. Given the continuous state space, there are some technicalities that need to be mentioned. For one thing, for an information structure π, one can change $\pi(s_i \mid \theta)$ for some $\theta \in \Theta_o$ without changing the ex-ante payoffs as long as Θ_o has measure 0. It should be understood later that if we claim that an optimal structure must have certain features, we have ignored such alternations. For another, in a similar fashion, one can add more signal realizations that have zero probability to be realized into an information structure without changing the ex-ante payoffs. For this reason, we assume that in all information structures considered below, all signal realizations can be realized with positive total probabilities.

3 Mixing Signal Realizations: An Example

In the canonical cheap talk game (Crawford and Sobel, 1982), Sender partitions the space into small intervals in an equilibrium, and Receiver only learns the interval that contains the true state. If this is still the case in an information control model, the analysis can be vastly simplified. Nevertheless, in this section, we provide a simple example to show that even with a simple prior such as Uniform$[0,1]$, a partitional structure may not be optimal.

As a preparation, we first rewrite the Receiver's ex-ante expected payoff. We can write (1) as

$$\begin{aligned}\mathbb{E}U^R(\pi) &= \sum_{i=1}^{N}\int_{\Theta}\left(2x_i\theta - x_i^2 - \theta^2\right) f(\theta)\pi(s_i \mid \theta)\mathrm{d}\theta \\ &= \sum_{i=1}^{N}\int_{\Theta}\left(2x_i\theta - x_i^2\right) f(\theta)\pi(s_i \mid \theta)\mathrm{d}\theta - \sum_{i=1}^{N}\int_{\Theta}\theta^2 f(\theta)\pi(s_i \mid \theta)\mathrm{d}\theta \\ &= \sum_{i=1}^{N}\int_{\Theta}\left(2x_i\theta - x_i^2\right) f(\theta)\pi(s_i \mid \theta)\mathrm{d}\theta - \int_{\Theta}\theta^2 f(\theta)\mathrm{d}\theta.\end{aligned}$$

Observe that in the last line, the second term does not depend on the information structure, so it can be omitted when we only aim to compare different structures. This also implies that what actually matters is the part $2x_i\theta - x_i^2$.

Consider the integral that depends on information structures. The first term is

$$\int_{\Theta} 2\theta x_i f(\theta)\pi(s_i \mid \theta)\mathrm{d}\theta = 2x_i w_i \int_{\Theta}\theta\frac{f(\theta)\pi(s_i \mid \theta)}{w_i}\mathrm{d}\theta = 2w_i x_i^2.$$

where

$$w_i \equiv \int_0^1 \pi(s_i \mid \theta) f(\theta)\mathrm{d}\theta$$

is the overall probability, or the *weight* of signal realization s_i is sent according to π.

The second term is

$$\int_{\Theta} -x_i^2 f(\theta)\pi(s_i \mid \theta)\mathrm{d}\theta = -x_i^2\int_{\Theta} f(\theta)\pi(s_i \mid \theta)\mathrm{d}\theta = -w_i x_i^2.$$

Substituting these results into (1),

$$\mathbb{E}U^R(\pi) = \sum_{i=1}^{N} w_i x_i^2. \tag{2}$$

The Receiver's goal is to choose an information structure that maximizes (2).

We now consider an example that shows mixing between different signal realizations in some states can improve the Receiver's ex-ante expected payoff. Let F be Uniform$[0,1]$, and $b = 0.126$. Notice that the bias is just slightly larger than $1/8$, implying that it is impossible to fit four partition cells while maintaining (IC). To see this, suppose x_i are all generated by partition cells $(a_{i-1}, a_i]$, with $a_0 = 0$ and $a_4 = 1$.[7] (IC) implies that

$$x_2 - x_1 = \frac{a_2 + a_1}{2} - \frac{a_1 + a_0}{2} \geqslant 2b,$$

where the first equality follows from uniform prior. This shows

$$a_2 - a_0 = a_2 \geqslant 4b.$$

[7]Throughout this paper, following the notation in Crawford and Sobel (1982), we will use a_i to denote partition points.

Similarly, since $x_4 - x_3 \geqslant 2b$, we also have

$$a_4 - a_2 = 1 - a_2 \geqslant 4b.$$

Together, we have $8b \leqslant 1$, which shows that if $b > 1/8$, a partitional structure with four posterior means cannot satisfy all (IC) constraints.

However, if one allows mixing signal realizations in some states, fitting four posterior means is still possible. For instance, we can consider the following structure:

- Partition cells $[0, a_1]$ and $[a_3, 1]$ induce the first and last posterior means, x_1 and x_4, respectively. Ideally, to fit four posterior means, the Receiver wants to push a_1 closer to 0 and a_3 closer to 1 compared with a partitional structure.
- In the middle region $[a_1, a_3]$, the states can send s_2 or s_3, inducing x_2 and x_3, respectively.
- For simplicity of the calculation, let $x_2 - x_1 = 2b$ and $x_4 - x_3 = 2b$.

This structure is shown in Figure 1.

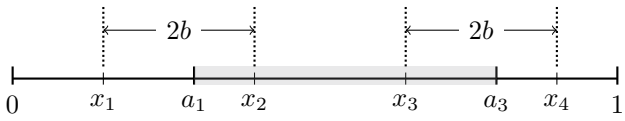


Figure 1 A class of information structures with mixing

One particular information structure that fits the description above is as follows.

- $a_1 = 0.244$, $a_3 = 0.756$, so that $x_1 = 0.122$ and $x_4 = 0.878$.
- For states between 0.244 and 0.4446, s_2 is sent for sure. For states between 0.5554 and 0.756, s_3 is sent for sure.
- For states between 0.4446 and 0.5554, both s_2 and s_3 are sent with positive probability. The probability of sending s_2 decreases linearly from 1 at $\theta = 0.4446$ to 0 at $\theta = 0.5554$.

This structure is visualized in Figure 2. The horizontal axis represents the states, the vertical axis represents the probabilities signals are sent at a state. Later, we will refer to parts of a structure similar involving mixing to the mixing within $[a_1, a_3]$ as *mixing regions*.

We can then use (2) to calculate the Receiver's payoff, which is approximately 0.328. The best partitional structure, since four-cell partition is not possible, is the three-cell evenly partitional structure that induces posterior means $1/6, 1/2, 5/6$ with equal probability. The Receiver's payoff in this case is $35/108 \approx 0.324$. Therefore, the best partitional structure is worse than the mixing structure considered above. We will revisit this example in subsection 5.3 after introducing some more general results regarding optimal information structures.

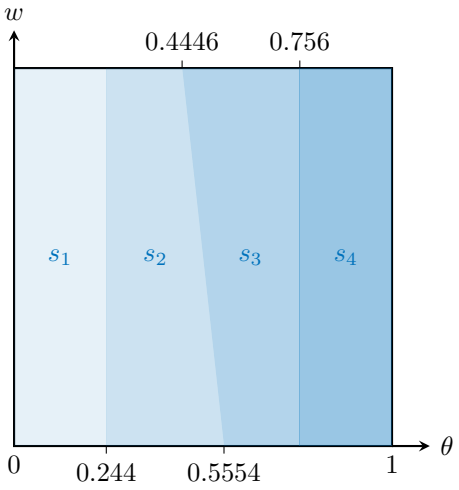


Figure 2 An example structure

4 The Integrated CDF Framework

The example above shows that it is not sufficient to consider partitional structures only. As mixing may lead to higher payoffs, we necessarily need to formally consider the Receiver's optimization problem in its full generality.

4.1 Feasibility

Gentzkow and Kamenica (2016) pointed out that not every distribution of posterior means $\{(x_i, w_i)\}$ such that $\sum w_i x_i = \mu$ can be generated by an information structure and provided a tool — the integrated CDFs — to identify distributions of posterior means that can be induced by information structures. Let G be some potential distribution of posterior means. The integrated CDF is

$$c(x) = \int_0^x G(t)\mathrm{d}t.$$

There are two extreme cases: the most informative information structure that reveals the true state, denoted by $\overline{\pi}$, and the least informative information structure that reveals no information at all, denoted by $\underline{\pi}$. The distribution of the posterior means of the former is just the prior F, while the latter is a degenerate distribution with the singleton support at μ. Therefore, the integrated CDFs are

$$c_{\overline{\pi}}(x) = \int_0^x F(t)\mathrm{d}t \quad \text{and} \quad c_{\underline{\pi}}(x) = \begin{cases} 0 & \text{if } 0 \leqslant x < \mu, \\ x - \mu & \text{if } \mu \leqslant x \leqslant 1. \end{cases}$$

Then G is feasible if and only if $c(x)$ is a convex function between $c_{\underline{\pi}}$ and $c_{\overline{\pi}}$ everywhere in $[0, 1]$.

In our case, we only need to consider information structures that induce finite posterior means. Given $\{(x_i, w_i)\}$, the integrated CDF is a $(N+1)$-segment piecewise linear function

$$c_\pi(x) = \begin{cases} 0 & \text{if } 0 \leqslant x < x_1, \quad \cdots \quad \text{Segment } 0 \\ w_1 x - w_1 x_1 & \text{if } x_1 \leqslant x < x_2, \quad \cdots \quad \text{Segment } 1 \\ \quad \vdots & \\ \left(\sum_{j=1}^{i} w_j\right) x - \sum_{j=1}^{i} w_j x_j & \text{if } x_i \leqslant x < x_{i+1}, \quad \cdots \quad \text{Segment } i \\ \quad \vdots & \\ x - \sum_{j=1}^{N} w_j x_j & \text{if } x_N \leqslant x \leqslant 1, \quad \cdots \quad \text{Segment } N. \end{cases} \tag{3}$$

Notice that the segments are numbered starting from 0. Segment 0 is constantly 0 for all integrated CDFs, so that the second segment, Segment 1, is the first "meaningful" segment.[8] Formally, Segment k is the line segment between $(x_k, c_\pi(x_k))$ and $(x_{k+1}, c_\pi(x_{k+1}))$, with $x_0 = 0$ and $x_{N+1} = 1$.

4.2 Connections between the Integrated CDF and the Objective Function

Following the observation that $c_{\overline{\pi}}$ is strictly above $c_{\underline{\pi}}$, it is reasonable to conjecture that the relative locations of the integrated CDF have implications on the informativeness of the information structures. This is indeed the case since the integrated CDFs are strongly related to the concept of *mean-preserving spread* (See, for example, Machina and Pratt (1997)). Here, by making a "spread", the signal becomes more informative, and at the same time, the integrated CDF becomes higher. This intuition then leads us to the idea that the area below the integrated CDF may be related to the Receiver's payoff.

Proposition 1. *The Receiver's expected utility is higher if the area below c_π is larger.*

The proof is algebraic and in Appendix A. This is a stronger result than mean-preserving spread since it does not require the integrated CDF of one distribution to be higher than the other pointwise. Notice that this means we can make yet another transformation of the objective function. Starting from (1), apart from the already simple algebraic expression in (2), we can also use the geometric result in Proposition 1 to compare two information structures.

[8] Observe that the last segment, which is just $x - \mu$, is also the same among all integrated CDFs. In other words, the piecewise linear integrated CDF must start and end on $c_{\underline{\pi}}$.

5 Features of Optimal Structures

The integrated CDF framework, in particular, Proposition 1, enables us to obtain some useful insights for characterizing the optimal information structures.

5.1 Graphical Observations

Geometrically, the Receiver's task is to find an $(N+1)$-piece piecewise linear convex function between $c_{\overline{\pi}}$ and $c_{\underline{\pi}}$ that has as large area below as possible, while keeping (IC) by separating each pair of posterior means at least $2b$ away. This naturally means that the Receiver should push the segments up so that they are tangent to $c_{\overline{\pi}}$ — the highest positions possible. A special case is $N_{\max}(b) \leqslant 3$, in which case tangency must be achieved at each segment of the integrated CDF. Formally,

Lemma 2. *In an optimal information structure, Segments* 1 *and* $N-1$ *must be tangent to* $c_{\overline{\pi}}$*. If at most two or three posterior means can be induced in all truth-telling information structure, then the integrated CDFs of all optimal information structures must be tangent to* $c_{\overline{\pi}}$ *at all segments.*

Proof. Here we consider the case that there can be at most three posterior means, which shows the idea. Since Segments 0 and 3 are fixed and tangent to $c_{\overline{\pi}}$ at $x = 0$ and $x = 1$, respectively, we only need to consider Segments 1 and 2. Suppose Segment 1 is not tangent to $c_{\overline{\pi}}$, as shown in the left panel of Figure 3. In Figure 3, the opaque lines are $c_{\overline{\pi}}$ (above) and $c_{\underline{\pi}}$ (below), and the solid line is the integrated CDF being examined.

Now consider the following modifications. Move x_1 to the left slightly, as shown in the right panel of Figure 3. For small enough change, the new Segment 1 is still under $c_{\overline{\pi}}$. The IC condition is preserved since the distance between x_1 and x_2 is larger, and finally the area below the integrated CDF is larger, implying that the ex-ante payoff of the Receiver is higher.

Similar argument can be made if Segment 2 is not tangent to $c_{\overline{\pi}}$, in which case one can move x_3 slightly to the right. □

When more posterior means are possible, we cannot claim that all segments are tangent to $c_{\overline{\pi}}$ due to IC constraints. Still, there are some observations we can make regarding segments that are not tangent to $c_{\overline{\pi}}$, if such segments can ever appear in an optimal structure. Consider an optimal information structure $\pi^\star$ with N posterior means. Let $c_{\pi^\star}$ be the corresponding integrated CDF. Moreover, we have the following two observations.

Lemma 3. *Let* π *be an optimal structure.* $c_{\pi^\star}$ *cannot have two consecutive segments not tangent to* $c_{\overline{\pi}}$*.*

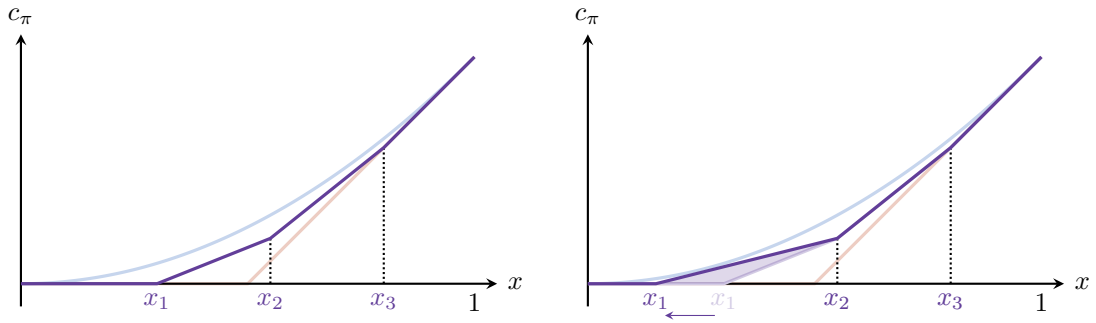


Figure 3 Proof of Lemma 2

Proof. See the left panel of Figure 4. Consider the modification that increases the vertical coordinate of $c_{\pi^\star}$ at x_i slightly. For small increases, the new $c_{\pi^\star}$ is still convex and under $c_{\overline{\pi}}$, yet the area below the new $c_{\pi^\star}$ is strictly larger. Notice that the (IC) condition is still preserved since the locations of x_i's are not changed under this modification. □

Lemma 4. *Let π be an optimal structure. If Segment i in $c_{\pi^\star}$ is not tangent to $c_{\overline{\pi}}$, (IC) must bind at Segment $i-1$ and $i+1$.*

Proof. See the right panel of Figure 4. Suppose $x_{i+2} - x_{i+1} > 2b$. Consider the modification that moves x_{i+1} slightly to the right. That is, the modification fixes the locations of x_{i-1}, x_i and x_{i+2} but raises the slope of Segment i, so that the intersection of Segment i and Segment $i+1$ is now higher and more to the right. Since the IC constraint between x_{i+1} and x_{i+2} is not binding, if the move is small, (IC) is not violated anywhere, and the area below the new $c_{\pi^\star}$ is strictly larger. □

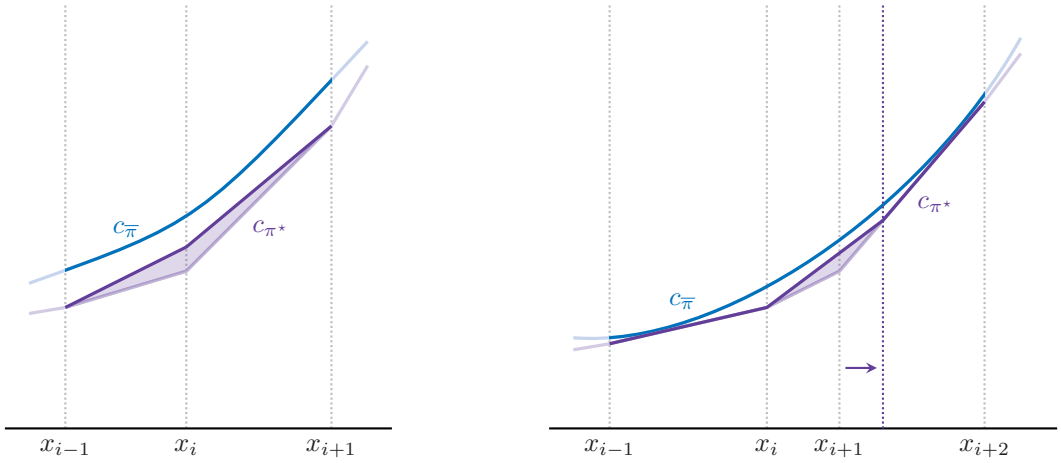


Figure 4 Proofs of the graphical observations

5.2 Partitions

The results above show that the integrated CDF of optimal structures are repeatedly tangent to the upper bound, $c_{\overline{\pi}}$. This leads to the question: what does tangency mean for an information structure? It turns out that tangency is closely related to a feature of the information structure being quasi-partitional.

Proposition 5. *Let π be a finite information structure. The following are equivalent:*

- *Segment i of the integrated CDF of π is tangent to $c_{\overline{\pi}}$ at $(m, c_{\overline{\pi}}(m))$.*
- *The signal realizations in π never cross over m: a signal realization is sent only from states below m or above m, but never both.*

An example is pictured in Figure 5 using the prior Uniform$[0, 1]$. The corresponding partitional structure is the evenly partitional structure, with the partition point at $1/3$ and $2/3$. The three posterior means are $x_1 = 1/6$, $x_2 = 1/2$ and $x_3 = 5/6$, respectively.

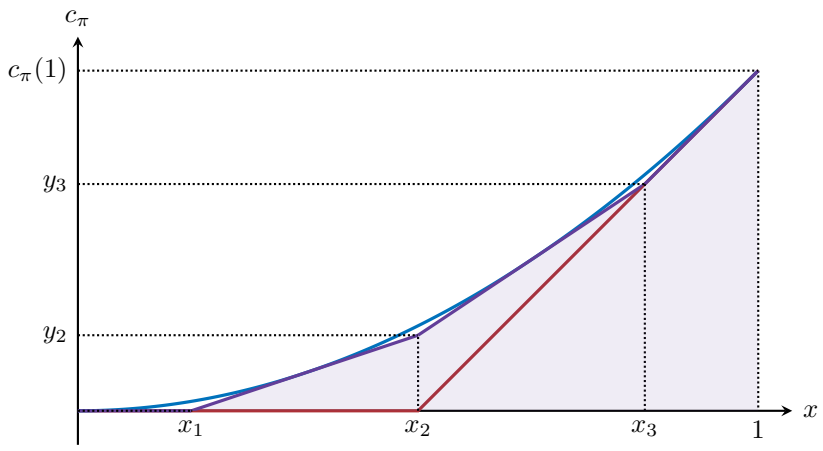


Figure 5 An example with three possible posterior means

The proof of Proposition 5 is in Appendix A. The intuition is that whenever there is a tangency point $(m, c_{\overline{\pi}(m)})$, one can "cut" the integrated CDF at $x = m$, so that the integrated CDF on $[0, m]$ and the integrated CDF on $[m, 1]$ both become a new qualified integrated CDF that represents the same information structure as before, restricted to the corresponding subinterval. Proposition 5 is the bridge between the graphical observations of the integrated CDF and the features of the information structure being represented. Using the graphical observations in Lemma 2, Lemma 3, and Lemma 4, together with Proposition 5, we have the following *partial characterization* of the optimal information structures.

Theorem 6. *For any optimal information structure of the Receiver:*

- *There are only two types of components in the structure: partition cells that each contain one posterior mean, or "mixing regions" that contain two posterior means each.*
- *If the optimal structure contains a mixing region that induces x_i and x_{i+1}, then the IC constraints between x_{i-1} and x_i, and between x_{i+1} and x_{i+2} must bind.*
- *Each posterior mean will not be induced by any state outside its cell or its mixing region.*
- *The optimal structure must start and end with partition cells.*

This result states that there are only two types of possible components in an optimal structure and indicates additional features regarding how the structure starts or ends, as well as how the structure behaves around mixing regions. The first feature corresponds to the graphical observation that there are no two consecutive segments of the integrated CDF of an optimal information structure that are both not tangent to $c_{\overline{\pi}}$. For example, if Segment $i-1$ and Segment i are both tangent (at a_i and a_{i+1}, respectively), then x_i is induced by the partition cell $[a_i, a_{i+1}]$. If Segment i of the integrated CDF is not tangent, then Segments $i-1$ and $i+1$ must be tangent, and x_i and x_{i+1} are induced by a mixing region. The second feature also corresponds to the observation that the IC constraint must bind around a non-tangent segment. The third feature is directly from Proposition 5. The last feature is an implication of Lemma 2.

We should note that the result that IC conditions are binding around mixing regions does not depend on prior distributions (although placing a mixing region in a particular place depends on the prior distribution.). Intuitively, mixing is inferior compared with partition cells by generating less clearer signal realizations; that is, the posterior means are less "spread". Without binding IC conditions around it, the Receiver would have some flexibility to reduce the extent of mixing and in turn improve the informativeness of the structure by increasing the extent of spread within the mixing region.

Using Lemma 2 and Theorem 6, we in particular, observe that if bias is relatively large and $N_{\max} \leqslant 3$, the optimal structure does not involve mixing.

Corollary 7. *If $b \geqslant 1/6$, then the optimal information structure must be a partitional structure.*

Another implication of Theorem 6, together with the observation that the Receiver's payoff in (2) only depends on $(w_i, x_i)_{i=1}^N$ is as follows.

Corollary 8. *For each optimal structure, there is a payoff-equivalent structure (hence also optimal) such that a deterministic signal realization is sent at each state.*

The proof is again relegated to Appendix A. To illustrate the idea, let's consider a mixing region $[\ell, h]$ that induces x_i and x_{i+1}. Suppose x_i is induced with weight w_i. Pick some $\tilde{\ell} \in [\ell, h]$ and find $\tilde{h}$ such that $\int_{\tilde{\ell}}^{\tilde{h}} f(\theta)\mathrm{d}\theta = w_i$. If we pick $\tilde{\ell} = \ell$, $[\tilde{\ell}, \tilde{h}]$ is the set of the lowest states within $[\ell, h]$ that has weight w_i. Therefore, when sending s_i in $[\tilde{\ell}, \tilde{h}]$, the induced posterior mean $\tilde{x}_i$ must

be weakly smaller than x_i. Similarly, if we pick $\tilde{\ell}$ such that $\tilde{h} = h$ and send s_i in $[\tilde{\ell}, \tilde{h}]$, these states are the highest states within $[\ell, h]$that has weight w_i, so that the induced posterior mean $\tilde{x}_i$ must be weakly larger than x_i. In turn, by the intermediate value theorem, there must be some $\tilde{\ell}$ in between such that exactly x_i is induced. Since we only have two signal realizations, what is left, $[\ell, \tilde{\ell}] \cup [\tilde{h}, h]$ must have weight w_{i+1} and induces x_{i+1}.

Of course, any structure that induces this distribution will be optimal, and there is no need to use a "linear" probability, or "straight line division" between two signal realizations involved in a mixing. For example, the following structure in Figure 6 is a four-signal structure such that when a state mixes between s_2 and s_3, they are mixed with equal probability. One can verify that this structure is payoff-equivalent to the structure in Figure 2.

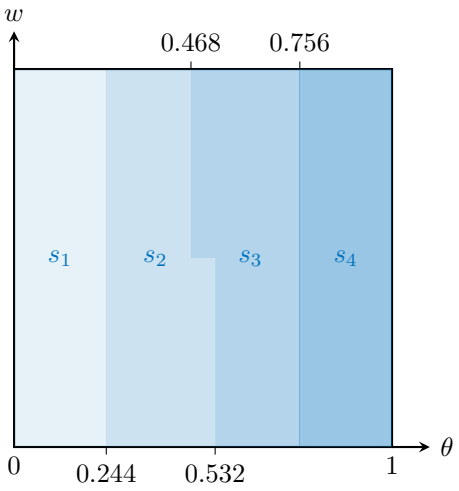


Figure 6 Another example of four-signal optimal structures

5.3 Four-Posterior-Mean Example

In this section, we show how the partial characterization of the optimal information structures from Theorem 6 can help in finding the optimal structure by revisiting the uniform prior example first introduced in section 3.

First, notice that the class of the structures introduced in section 3 indeed satisfy all the features in Theorem 6. Also, since the mixing regions cannot be placed at the beginning or the end of the support, the structure is also the *only* possible class of structure that is optimal and involves mixing.

Since the structure begins and ends with partition cells, the total probability weights that s_1 and s_4 are sent (or equivalently, that x_1 and x_4 are induced) are

$$w_1 = 2x_1 \quad \text{and} \quad w_4 = 2(1 - x_4).$$

To calculate the other two weights, the following two relations are used:

$$\sum_{i=1}^{4} w_i = 1 \quad \text{and} \quad \sum_{i=1}^{4} w_i x_i = \frac{1}{2}.$$

Substituting w_1 and w_4 back and solving the system of equations to get

$$w_2 = \frac{\frac{1}{2} - x_3 - 2x_1(x_1 - x_3) - 2(1 - x_4)(x_4 - x_3)}{x_2 - x_3},$$
$$w_3 = \frac{\frac{1}{2} - x_2 - 2x_1(x_1 - x_2) - 2(1 - x_4)(x_4 - x_2)}{x_3 - x_2}.$$

Further substitute $x_2 = x_1 + 2b$ and $x_3 = x_4 - 2b$ to get

$$w_2 = \frac{(2x_4 - 2x_1 - 1)(1 - 2x_1 - 4b)}{2(x_4 - x_1 - 4b)},$$
$$w_3 = \frac{(2x_4 - 2x_1 - 1)(2x_4 - 4b - 1)}{2(x_4 - x_1 - 4b)}. \tag{4}$$

The algebra here shows that the Receiver in fact only needs to consider two parameters: x_1 and x_4. All other parameters in the optimization can be written as functions of x_1 and x_4. This is due to the fact that the optimal information structure is already largely pinned down by Theorem 6.

The Receiver's optimization problem is to find the optimal four-signal structure described above and to compare it with the optimal three-signal structure, which is the three-cell evenly partitional structure.

Theorem 9. *Let $1/8 < b < 1/6$. There exists $\bar{b} \in (1/8, 1/6)$ such that if $b < \bar{b}$, the Receiver's optimal information structures is a four-signal structure that satisfies:*

- *The incentive compatibility constraint is binding everywhere.*
- *The first posterior mean is $x_1 = \frac{1}{2} - 3b$.*
- *Both x_1 and x_4 are induced by partition cells, while x_2 and x_3 are induced by a mixing region.*

If $b > \bar{b}$, the optimal structure is the three-cell evenly partitional structure. Both structures are optimal if $b = \bar{b}$.

The proof uses arguments involving standard Kuhn-Tucker conditions and is in Appendix A. If we set $b = 0.126$, the optimal structure is the one presented in section 3.

It may be helpful to consider why mixing could be useful in more detail. For now, let's assume that IC is binding everywhere. For a given pair (x_2, x_3) such that $x_3 = x_2 + 2b$, if it is generated by a symmetric[9] mixing region, then mixing feasibility — x_2 and x_3 can be generated by a mixing

[9] In the sense that $x_2 - a_1 = a_3 - x_3$ and the two posterior means are induced with the same weight. There is no reason to consider asymmetric mixing region given the uniform prior, although we did not impose symmetry when showing Theorem 9. We also did not impose that IC constraints are binding everywhere.

region, meaning that x_2 and x_3 cannot be too far away — implies $x_2 - a_1 > b$ and $a_3 - x_3 > b$. This in turn shows that $a_1 - x_1 < b$ and $x_4 - a_3 < b$, so that the first and the last cells are smaller. This shows that (i) the mixing region itself does not save space; instead, it captures more space (size larger than $4b$) than two partition cells (total size equal to $4b$); and (ii) the mixing region fits more signal realizations by shifting the conditional expectations of the state after signals in the mixing region closer to each other, away from the state boundaries.

If $b = 1/8$, the structure is the four-cell even-partitional structure. When b is slightly larger, as shown in Figure 1 or Figure 6 for $b = 0.126$, the optimal four-signal structures are very close to the four-cell evenly partitional structure. Intuitively, the partitional structures with more cells minimize the conditional variances, and the even partitions then guarantee all states are valued equally, which is ideal given the uniform distribution. Therefore, the Receiver prefers the four-cell evenly partitional structure if possible. Yet when $b > 1/8$, the IC constraints rule out the four-cell evenly partitional structure, so that the best the Receiver can do is to mimic the four-cell evenly partitional structure by adding minimal mixing in the middle and slightly squeezing the boundary cells. However, if b is significantly larger and closer to $1/6$, the mixing structures resemble to the four-cell evenly partitional structure less, meaning that the Receiver has to mix heavily in the middle and reduce the size of the boundary cells by a lot. The mixing and unevenness then lead to worse payoffs than having one fewer signal realizations, and the three-cell evenly partitional structure indeed performs better.

The same logic applies when b is slightly smaller than $1/8$. In this case, a five-signal structure is available, but it is rather uneven. The Receiver would like to use the four-cell evenly partitional structure instead. This is shown in Figure 7. As b further decreases, the Receiver would switch to a five-signal structure.

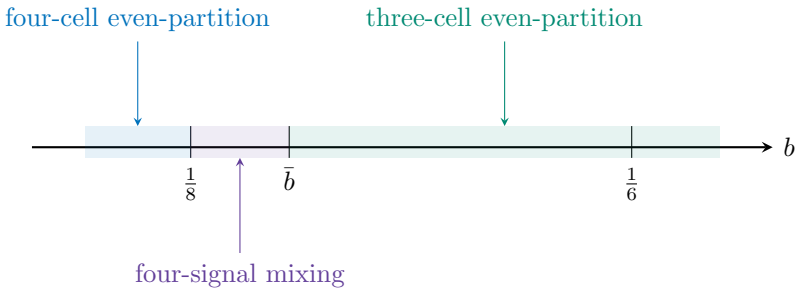


Figure 7 Optimal information structure of the Receiver for different biases

Theorem 9 also shows in general how one can use Theorem 6 to solve optimal information structures. Since the optimal structure can only contains two types of components, the Receiver should first consider the possibilities of numbers and locations of mixing regions (i.e., the classes of structures). In the uniform example, when $N_{\max} = 4$, there is at most 1 mixing region that must

locate in the middle. For larger $N_{\max}$, the Receiver could use more mixing regions or place the mixing regions in different locations. The way that Theorem 6 helps simplify the Receiver's problem is by (i) reducing the possibilities of structures that need to be considered, and (ii) reducing the number of choice variables in the optimization problem.

6 Extensions

The approach developed in the previous sections relies on two features of the baseline environment. First, the Sender's bias is assumed to be constant across states, which implies that incentive compatibility depends only on the distribution of posterior means. Second, the prior distribution is assumed to admit a continuous density, which suggests a smooth upper bound $c_{\overline{\pi}}$ of the integrated CDF so that tangency can be achieved relatively easily. In Section 6.1, we consider two extensions that show both conditions can be relaxed, and the integrated CDF approach continues to be applicable under weaker assumptions.

In addition, the approach can help solve information design problems where the relevant constraints are different than in our base model. In Section 6.2, we provide two examples where the Sender faces exogenous capacity constraints on information transmission.

6.1 Technical Extensions

We first discuss how the integrated CDF approach can be applied under weaker assumptions. More specifically, in Section 6.1.1, we allow the Sender's bias to vary with the underlying state. A challenge in this extension is that the posterior mean distribution no longer fully summarizes the Sender's incentives: different information structures inducing the same distribution of posterior means may generate different incentive constraints. We show that assuming that the bias function is affine restores the key property needed for the integrated CDF approach, and the main characterization result continues to hold. In Section 6.1.2, we consider priors with finitely many mass points, including finite state spaces as a special case. We show that the integrated CDF approach remains applicable, and the optimality condition in Theorem 6 can be adapted to priors with mass points. In addition, the presence of mass points introduces new possibilities, including partial or full revelation of the mass points.

6.1.1 State-dependent Bias

Suppose that the homogeneous bias b is replaced by a bias function $b(\theta)$, where $b(\theta) > 0$ for all θ. Conditional on receiving signal s_i, the Sender does not observe the realized state. Instead, the Sender evaluates the reporting decision using the entire posterior distribution induced by the

signal. Under state-dependent bias, the Sender now considers the average bias over the posterior distribution.

Formally, suppose that the bias function $b(\cdot)$ allows for well-defined expectations. Let x_i denote the posterior mean associated with signal s_i, and define the expected bias conditional on signal s_i as

$$\beta_i = \mathbb{E}[b(\theta)|s_i].$$

The Sender's expected utility from reporting action a, conditional on s_i, can be written as

$$\begin{aligned} -\mathbb{E}[(a-\theta-b(\theta))^2 \mid s_i] &= -\mathbb{E}[a^2 - 2a\theta - 2ab(\theta) \mid s_i] - \mathbb{E}[(\theta+b(\theta))^2 \mid s_i] \\ &= -[a^2 - 2a(x_i+\beta_i)] - \mathbb{E}[(\theta+b(\theta))^2 \mid s_i] \end{aligned}$$

Since the second term does not depend on the reported action, the Sender's preferred action after receiving s_i is

$$x_i + \beta_i.$$

Therefore, the incentive compatibility constraint between two adjacent posterior means becomes

$$x_{i+1} - x_i \geq 2\beta_i = 2\mathbb{E}[b(\theta)|s_i].$$

Unlike the baseline model, the Sender's incentive constraint depends not only on the posterior mean x_i, but also on the entire posterior distribution of states associated with the signal. In particular, two information structures that induce the same distribution of posterior means may generate different values of $\mathbb{E}[b(\theta)|s_i]$, and therefore may differ in whether they satisfy incentive compatibility.

This additional dependence creates difficulties for the integrated CDF approach. The local perturbations used in Lemma 2, Lemma 3, and Lemma 4 modify the distribution of posterior means while preserving the feasibility of the integrated CDF. Under constant bias, these perturbations leave the bias terms unchanged, so their effects on IC can be determined directly from the movements in posterior means because the bias term is identical across signals. With state-dependent bias, however, a local change in the integrated CDF may require a change in the underlying state-to-signal assignment. Such a change can have non-local effects on the conditional distributions associated with multiple signals. For example, consider the local perturbation used in the proof of Lemma 4, which is shown in the right panel of Figure 4. Here, x_i and x_{i+1} are generated from a mixing region. The local perturbation slightly increases x_{i+1} by rotating the segment between x_i and x_{i+1} of c_π. This means that the signals s_i, s_{i+1} now can be assigned to different states in the mixing region, which in turn changes β_i and β_{i+1}. This implies the following two IC constraints

$$x_{i+1} - x_i \geqslant 2\beta_i \quad \text{and} \quad x_{i+2} - x_{i+1} \geqslant 2\beta_{i+1}$$

do not necessarily hold anymore. To further see the tension, we focus on the first IC constraint regarding x_{i+1} and x_i. A small increase in x_{i+1} increases the distance between x_i and x_{i+1}, which relaxes the IC constraint in the base model. But with state-dependent bias, the bias term β_i may also change due to the reassignment between states and signals in the mixing region. Consequently, depending on the sizes of the two small changes, the IC constraint is no longer guaranteed to be preserved. This suggests that to apply the graphical framework to the problems with state-dependent biases, additional assumptions are required to control the size of the changes of β_i's under local perturbation of x_i's.

Assumption 1 (Affine Bias)**.** *Suppose the bias function b takes the following form,*

$$b(\theta) = \alpha + \rho\theta,$$

where $\alpha > \max\{0, -\rho\}$ and $\rho \geqslant -1/2$.

Given this assumption,

$$\beta_i = \mathbb{E}\left[b(\theta) \mid s_i\right] = \mathbb{E}\left[\alpha + \rho\theta \mid s_i\right] = \alpha + \rho x_i = b(x_i).$$

Notice that the size of the bias is an affine function of the posterior mean. For an arbitrary bias function, the problem is that the expected bias β_i depends on the entire posterior distribution of states after receiving s_i. The local perturbation could generate non-local changes of the posterior distribution, and the bias need not change proportionally to the change of posterior mean. With affine bias, the expected bias β_i is now anchored at the posterior mean, so that a local change in the posterior mean implies a local change in the average bias. The condition $\rho \geqslant -1/2$ ensures that the bias does not increase too fast as the posterior mean decreases, and the condition on α ensures that the bias is positive everywhere.

Under affine bias, the incentive compatibility constraint becomes

$$x_{i+1} - x_i \geqslant 2b(x_i). \tag{5}$$

The integrated CDF again contains all the information necessary to evaluate the Sender's incentives, restoring the key property behind the integrated CDF approach.

Theorem 10. *Suppose the Sender's bias function satisfies Assumption 1. Then the characterization in Theorem 6 continues to hold, with the IC constraints replaced by (5).*

The proof is in Appendix A, which verifies that Lemma 2, Lemma 3, and Lemma 4 still hold under Assumption 1. To see the role of the condition $\rho \geqslant -1/2$, consider the local perturbation in Figure 3. With homogeneous bias, when decreasing x_1, the distance between x_1 and x_2 increases, which should relax the IC constraint $x_2 - x_1 \geqslant 2b$. With state-dependent affine bias, however,

the right hand side is replaced by $2b(x_1)$, so that if $b(\theta)$ decreases in θ, the local perturbation in Figure 3 relaxes the IC from the LHS but tightens the IC from the RHS. The condition $\rho \geqslant -1/2$ implies that $b(\theta)$ cannot decrease in θ too fast, ensuring that the IC is indeed relaxed under such local perturbations.

6.1.2 Priors with Mass Points

Here we return to assuming that the bias $b > 0$ is constant, but we allow for the possibility that the prior distribution F may admit finitely many mass points. The direct challenge is that the integrated CDF function $c_{\overline{\pi}}$ of the perfect information is no longer smooth. Instead, at a state where there is a mass point, $c_{\overline{\pi}}$ has a kink, so ordinary differentiable tangency must be replaced by supporting contact. In this section, we first discuss how the integrated CDF corresponds to the information structure, especially when the integrated CDF of an information structure makes contact with $c_{\overline{\pi}}$ at a kink. We then provide an adapted version of Theorem 6 that characterizes the behaviors of the integrated CDF graphs of optimal information structures around the states with mass points.

To begin with, note that the main results of Gentzkow and Kamenica (2016) hold if the prior distribution has mass points. Because of this, all of our previous observations continue to hold, and what remains is to interpret the contact point of c_π and $c_{\overline{\pi}}$ at a state where there is a mass point.

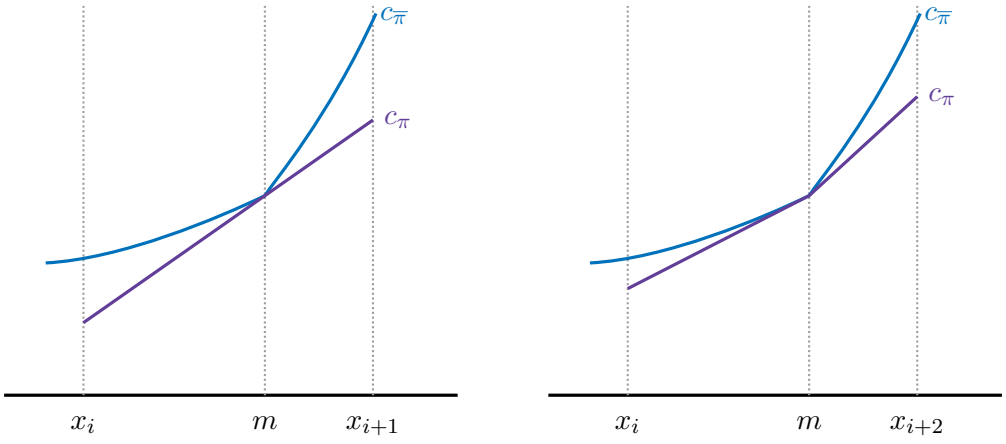


Figure 8 Contact between c_π and $c_{\overline{\pi}}$ at a mass point

Figure 8 illustrates the two cases where $c_{\overline{\pi}}$ and c_π make contact at the state m where the prior distribution has a mass point. The mass point appears as a kink on $c_{\overline{\pi}}$. In the left panel, one segment of c_π, Segment i, touches $c_{\overline{\pi}}$ from below at m. In the right panel, the mass point happens to be a boundary point between two segments of c_π. In other words, Segment i touches $c_{\overline{\pi}}$ from below at m on the left side, and Segment $i+1$ touches $c_{\overline{\pi}}$ from below at m on the right side. This

suggests m itself is also a posterior mean (i.e., x_{i+1}).

We first extend Proposition 5 and show that such contact points are also partition points of the information structure, just like tangency points.

Proposition 11. *Let m be the state where the prior distribution has a mass point. Suppose that an information structure associated with an integrated CDF function c_π has a contact point at m with $c_{\overline{\pi}}$. Then m is a partition point of the information structure; that is, signals sent at states below m are not sent at states above m, and vice versa.*

The proof is in the Appendix A. Proposition 11, together with Proposition 5, indicates that there is a partition point in the information structure whenever $c_\pi = c_{\overline{\pi}}$, whether this point is a tangency point or a contact point at a kink.[10] For the case in the left panel of Figure 8, m itself is not a posterior mean, so that the probability mass of m is divided by the two neighboring bi-pooling regions (cells or mixing regions), and the slope of Segment i determines how much probability mass is received by each region. More specifically, by (3), the slope of Segment i is the total weights for the first i signals, $\sum_{j=1}^{i} w_j$, which is a value in between the left derivative of $c_{\overline{\pi}}$, i.e., $F(m^-)$, and the right derivative of $c_{\overline{\pi}}$, i.e., $F(m)$. The probability mass of m assigned to the left bi-pooling region is then

$$\sum_{j=1}^{i} w_j - F(m^-),$$

and the remaining mass is assigned to the right region. In the extreme case, if Segment i is tangent to $c_{\overline{\pi}}$ at m from the left, then all probability mass of m is assigned to the right region. Conversely, if Segment i is tangent to $c_{\overline{\pi}}$ at m from the right, then all probability mass of m is assigned to the left region. For the case in the right panel of Figure 8, m itself is a posterior mean, meaning that there is a signal s_{i+1} that reveals state m. Similar to the argument above, $\sum_{j=1}^{i} w_j - F(m^-)$ is the probability mass of m assigned to the left region, $F(m) - \sum_{j=1}^{i+1} w_j$ is the probability mass of m assigned to the right region, and the remaining mass is assigned to signal s_{i+1} to reveal state m. These characterizations continue to hold if the mass point is at state 0 or state 1.

We now extend the partial characterization of the optimal information structures to priors that have finitely many mass points and otherwise satisfy the regularity conditions imposed in the baseline model. If a mass point lies strictly inside a component of an information structure, where a component is either a partition cell or a mixing region, it creates no additional complications for the integrated CDF approach. In particular, the cutoff points surrounding that component lie in smooth portions of $c_{\overline{\pi}}$, and the local perturbations underlying Theorem 6 remain applicable. It

[10] Proposition 11 does not rely on the assumption of a finite number of mass points. Nevertheless, the local perturbation of the graphical framework requires that at each mass point there is a (small) neighborhood that does not contain another mass point. For simplicity of the analysis, we maintain the assumption of finite mass points throughout.

is therefore sufficient to consider the cases in which a mass point is itself a cutoff point. We will consider the cases where a mass point is in $(0,1)$, and where a mass point is on the boundary, i.e., states 0 or 1.

Theorem 12. *Let π be an optimal information structure and m be a mass point of the prior.*

1. *(Interior mass point) Suppose that $c_\pi(m) = c_{\overline{\pi}}(m)$, and let $x_i < m < x_j$ be the posterior means immediately adjacent to m. Suppose that $m - x_i \geqslant 2b$ and $x_j - m \geqslant 2b$. If $F(m^-) < c'_\pi(m^-)$, then $x_i - x_{i-1} = 2b$ whenever x_{i-1} exists. If $c'_\pi(m^+) < F(m)$, then $x_{j+1} - x_j = 2b$ whenever x_{j+1} exists.*
2. *(Boundary mass point)*
   - *(Boundary mass point at state 0) Let $p_0 = F(0) > 0$, and let $r_0 = c'_\pi(0^+)$ be the amount of the mass point that is perfectly revealed. Let $x_1 < x_2$ be the two posterior means closest to state 0. If $0 < r_0 < p_0$, then $x_2 - x_1 = 2b$ whenever x_2 exists. If $0 < r_0 \leqslant p_0$ and $c_\pi < c_{\overline{\pi}}$ for all $x \in [x_1, x_2)$, then $x_1 = 2b$.*
   - *(Boundary mass point at state 1) Let $p_1 = 1 - F(1^-) > 0$, and let $r_1 = 1 - c'_\pi(1^-)$ be the amount of the mass point that is perfectly revealed. Let $x_{N-1} < x_N$ be the two posterior mean approach means closest to state 1. If $0 < r_1 < p_1$, then $x_N - x_{N-1} = 2b$. If $0 < r_1 \leqslant p_1$, and $c_\pi < c_{\overline{\pi}}$ for all $x \in (x_{N-1}, x_N]$, then $1 - x_N = 2b$.*

The proof is in the Appendix A. Theorem 12 highlights a distinctive role of mass points in the information-control problem. Since the Receiver benefits from more informative signals, she would ideally like to reveal the state perfectly. Under an atomless prior, separately revealing a single state affects only a zero-probability event and therefore provides no ex-ante payoff gain. By contrast, when the prior has a mass point, revealing that state allows the Receiver to implement the first-best action with positive probability. This benefit may justify using less informative signals for nearby states. Mass points can therefore serve as natural anchors around which the Receiver constructs the information structure, even when they are surrounded by states occurring with positive density.

This opportunity is not costless. Introducing a mass point as a posterior mean requires every neighboring posterior mean to remain at least $2b$ away. Revealing more of the atom may therefore tighten incentive constraints and reduce the amount of information that can be transmitted about nearby states. The Receiver trades off the direct benefit of perfectly revealing the atom against the resulting loss of flexibility elsewhere. The necessary conditions in Theorem 12 reflect this tradeoff: whenever some of the atom remains pooled despite being located at a cutoff point, a nearby binding incentive constraint generally prevents the Receiver from revealing more of it. Thus, the theorem identifies conditions under which the Receiver has exhausted all locally feasible gains from increasing the revelation probability of a mass point.

6.2 Capacity Constraints

We will next show that the integrated CDF approach can be useful in information design problems in which the designer faces not the incentive compatibility constraints of our base model, but exogenous capacity constraints. Such constraints are common in engineering problems (see Gray and Neuhoff (1998) for a survey). Below we consider the cases when there is an upper bound on the number of signals available for communication, and when there is an upper bound on the entropy of the information structure. In both problems we assume that the agent has perfectly aligned preference with the information designer, so the only relevant constraints stem from bounded capacity.

6.2.1 Upper Bound on the Number of Signals

Suppose that the designer can only include at most K signals in any information structure, where K is finite. An easy argument establishes that the optimal information structure is partitional using the integrated CDF approach.

Proposition 13. *The optimal information structure is partitional.*

Proof. Suppose not. Proposition 5 then suggests that the integrated CDF induced by the optimal structure must have a piece that is not tangent to $c_{\overline{\pi}}$, as shown in Figure 9. But then the designer can make a slight parallel shift of the non-tangent piece and achieve a strictly better payoff, which is a contradiction. □

Quadratic Utility Example. Having established that the optimal information structure is partitional, we can study the problem further by considering, for example, what specific features an optimal information structure exhibits with a specific utility function. Here, we consider quadratic utilities. Since the optimal structure is partitional, we can focus on K-cell partitional structures characterized by the partition points

$$0 = a_0 < a_1 < \cdots < a_{K-1} < a_K = 1.$$

Under quadratic utility, (2) gives the designer's objective function, which is

$$\sum_{k=1}^{K} \frac{\left[\int_{a_{k-1}}^{a_k} \omega f(\omega) \mathrm{d}\omega\right]^2}{\int_{a_{k-1}}^{a_k} f(\omega) \mathrm{d}\omega}. \tag{6}$$

For a generic $k = 1, \cdots, K-1$, there are only two terms in (6) containing a_k. That is,

$$\frac{\left[\int_{a_{k-1}}^{a_k} \omega f(\omega) \mathrm{d}\omega\right]^2}{\int_{a_{k-1}}^{a_k} f(\omega) \mathrm{d}\omega} + \frac{\left[\int_{a_k}^{a_{k+1}} \omega f(\omega) \mathrm{d}\omega\right]^2}{\int_{a_k}^{a_{k+1}} f(\omega) \mathrm{d}\omega}.$$

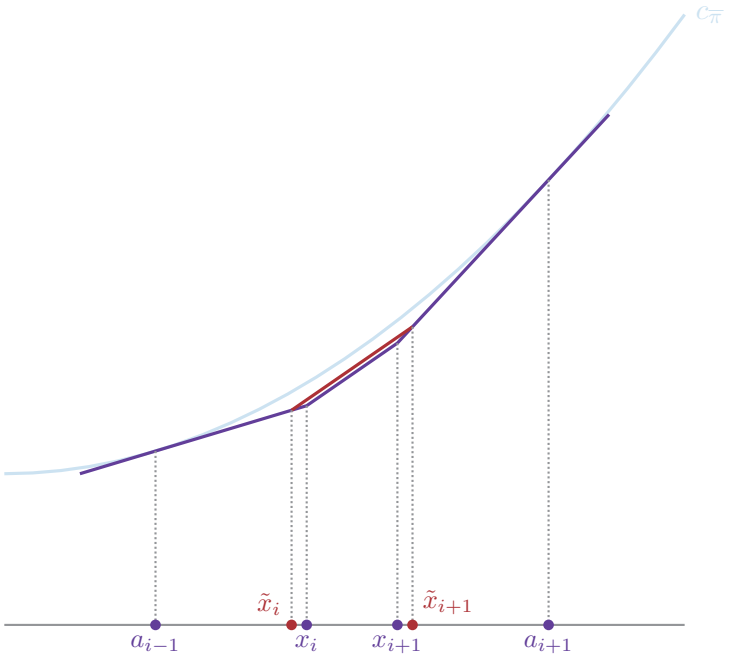


Figure 9 Parallel shift

Since the density is assumed to be positive everywhere, the first order condition can be simplified to

$$a_k = \frac{x_k + x_{k+1}}{2}. \tag{7}$$

This is summarized as follows.

Proposition 14. *Suppose that the Sender and the Receiver have the same utility function $-(y-\theta)^2$. In an optimal structure, the partition points must be the average of the two neighboring posterior means.*

In the case of quadratic utility and uniform prior, (7) simply gives the even partitional structures as expected. However, we should point out that under general prior distributions, the optimal structure does not assign equal weights to each cell, nor does it keep the conditional variances equalized among all the cells.[11]

6.2.2 Upper Bound on Information Entropy

A different type of constraint on the information structure that might apply to a Sender is a cap on its entropy, a commonly used measure of the informativeness of the structure. This can be

[11] As an example, consider the prior density $f(\theta) = 2 - 2\theta$, and let $K = 2$. The optimal partition point is approximately 0.382, and the weights of the two cells are approximately 0.618 and 0.382, respectively. The conditional variances are approximately 0.012 and 0.021, respectively.

the case if there is a fixed budget for establishing the information structure, and more informative structures are more expensive.

For an information structure that induces (discrete) distribution of posterior means $(w_k, x_k)_{k=1}^K$, the information entropy is defined by[12]

$$E(\pi) = -\sum_{k=1}^{K} w_k \ln w_k.$$

Consider the problem where the designer chooses an information structure π to maximize the expected utility under the induced posterior distribution subject to the constraint

$$E(\pi) \leqslant E. \tag{8}$$

Proposition 15. *The optimal information structure is partitional. Also, the entropy constraint* (8) *must bind in an optimal structure.*

To see that the optimal structure is partitional, recall that Segment i of the integrated CDF has the expression

$$\left(\sum_{j=1}^{i} w_j\right) x - \sum_{j=1}^{i} w_j x_j,$$

so the slope of the integrated CDF only depends on the weights of the signals. Also notice that the entropy of an information structure, given that the number of signals are fixed, also only depends on the weights of each signal. Therefore, as long as the slope of the integrated CDF are not altered, the entropy does not change. Then, the parallel shift argument in the proof of Proposition 13 applies again: if a segment is not tangent to the upper bound of the integrated CDF, the Sender can always apply a small enough upwards parallel shift and increase the expected payoff without changing the entropy of the information structure. For the proof that the entropy constraint must bind, see Appendix A. The idea is also simple: partitioning an existing cell always provides more information and increases the entropy. The entropy capacity E in (8) then acts as a budget constraint: if the constraint does not bind, the designer can always partition one existing cell further, leading to a higher payoff.

Quadratic Utility Example. As we now know that the optimal structure is partitional, we again consider the special case where the utility function is quadratic.

[12]For the ease of notation, we adopt the natural logarithm. Changing the base of the logarithm will not alter the qualitative results below. Also, since the definition of differential entropy (or continuous entropy) is known to be not fully compatible with the discrete version, we will rule out any information structure that induce truth-telling in a non-trivial subset of the state space by setting the entropy of such information structure to be $+\infty$, which is the limit of the entropy of finer and finer partitional structures.

The Lagrangian is

$$\mathcal{L} = \sum_{k=1}^{K} \frac{\left[\int_{a_{k-1}}^{a_k} \omega f(\omega) \mathrm{d}\omega\right]^2}{\int_{a_{k-1}}^{a_k} f(\omega)\mathrm{d}\omega} + \lambda \left[E + \sum_{k=1}^{K} \int_{a_{k-1}}^{a_k} f(\omega)\mathrm{d}\omega \ln\left(\int_{a_{k-1}}^{a_k} f(\omega)\mathrm{d}\omega\right)\right].$$

The first order condition with respect to a_k simplifies to (again using $f(\cdot) > 0$ everywhere)

$$a_k = \frac{x_k + x_{k+1}}{2} + \lambda \frac{\ln w_k - \ln w_{k+1}}{x_{k+1} - x_k}. \tag{9}$$

Comparing this with (7), i.e., the optimal a_k when the only constraint is the number of cells, the first term of (9) is exactly (7), while the second term adjusts the partition points to satisfy the entropy constraint. Due to the fact that (8) always binds in optimum, $\lambda > 0$. Since $x_{k+1} > x_k$ by construction, the sign of the second term is determined by the sizes of w_k and w_{k+1}.

However, the optimal structure can be more easily solved by reformulating the objective function. Importantly, since the values of posterior means do not enter the entropy, the problem can be further simplified if the conditional variances depend on the weights of the cells only. This is satisfied, for example, by a uniform prior. Assume that the prior is uniform for the remainder of this section, so that the ex-ante expected payoff is

$$-\frac{1}{12}\sum_{i=1}^{K} w_i^3.$$

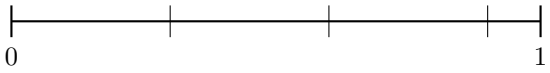
We can also without loss consider the upper bound of the entropy takes the form of $\ln(K+\varepsilon)$ for some $\varepsilon \in [0,1)$. Not only $\{[\ln K, \ln(K+1))\}_K$ cover $\mathbb{R}_+$, $\ln K$ is also the entropy of the K-cell even partitional structure. That is, for each K, we know that the optimal structure should perform at least as well as the K-cell evenly partitional structure.

Proposition 16. *Suppose that the prior distribution is uniform on* $[0,1]$*, and the designer's payoff function is quadratic. If the upper bound of the entropy is* $\ln K$ *for some* $K = 1, 2, 3, \cdots$*, the optimal structure is a* K*-cell evenly partitional structure. If the upper bound of the entropy is* $\ln(K+\varepsilon)$ *for some* $\varepsilon \in (0,1)$*, the optimal structure is a* $(K+1)$*-cell partitional structure such that:*

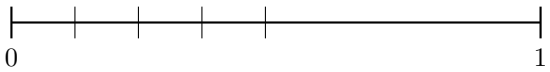
- *All but one cells share the same cell size.*
- *The remaining one cell has a smaller cell size than the other cells.*

The proof is in the Appendix A. To show this result, we start by establishing the standard Lagrangian for a generic k-cell structure (where k is at least K). The first order condition indicates that the optimal structure can have at most two cell sizes, and all but one cell must share the same size. We can then focus on the structures with this feature. It is easy to check that with $\varepsilon \in (0,1)$, there exists a $(K+1)$-cell structure such that it contains K equi-length cells and one additional smaller cell. However, for any structure with more cells (and the aforementioned feature),

the equal-length cells must be smaller than the one remaining cell (which holds for $(K+1)$-cell structures when $\varepsilon = 0$). One example with $K = 3$ and $\varepsilon \in (0, 1)$ is shown in Figure 10. By further checking the second order condition, if the structure is at least locally optimal, the cell size shared by all but one cells must be at least as large as the size of the remaining one cell. This rules out the possibility that the optimal structure has $(K+2)$ or more cells when $\varepsilon \in (0, 1)$, or the optimal structure has $(K+1)$ or more cells when $\varepsilon = 0$.



(a) A Four-Cell Structure



(b) A Five-Cell Structure

Figure 10 Fitting one more cell

References

Attila Ambrus, Eduardo M Azevedo, and Yuichiro Kamada. Hierarchical cheap talk. *Theoretical Economics*, 8(1):233–261, 2013.

Itai Arieli, Yakov Babichenko, Rann Smorodinsky, and Takuro Yamashita. Optimal persuasion via bi-pooling. *Theoretical Economics*, 18(1):15–36, 2023.

David Austen-Smith. Strategic transmission of costly information. *Econometrica: Journal of the Econometric Society*, pages 955–963, 1994.

David Blackwell. Equivalent comparisons of experiments. *The annals of mathematical statistics*, pages 265–272, 1953.

Vincent P Crawford and Joel Sobel. Strategic information transmission. *Econometrica: Journal of the Econometric Society*, pages 1431–1451, 1982.

Piotr Dworczak and Giorgio Martini. The simple economics of optimal persuasion. *Journal of Political Economy*, 127(5):1993–2048, 2019.

Paul E Fischer and Phillip C Stocken. Imperfect information and credible communication. *Journal of Accounting Research*, 39(1):119–134, 2001.

Matthew Gentzkow and Emir Kamenica. A rothschild-stiglitz approach to bayesian persuasion. *American Economic Review*, 106(5):597–601, 2016.

Robert M. Gray and David L. Neuhoff. Quantization. *IEEE transactions on information theory*, 44(6):2325–2383, 1998.

Jerry R Green and Nancy L Stokey. A two-person game of information transmission. *Journal of economic theory*, 135(1):90–104, 2007.

Maxim Ivanov. Communication via a strategic mediator. *Journal of Economic Theory*, 145(2): 869–884, 2010a.

Maxim Ivanov. Informational control and organizational design. *Journal of Economic Theory*, 145 (2):721–751, 2010b.

Maxim Ivanov. Optimal monotone signals in bayesian persuasion mechanisms. *Economic Theory*, 72(3):955–1000, 2021.

Emir Kamenica and Matthew Gentzkow. Bayesian persuasion. *American Economic Review*, 101 (6):2590–2615, 2011.

Andreas Kleiner, Benny Moldovanu, and Philipp Strack. Extreme points and majorization: Economic applications. *Econometrica*, 89(4):1557–1593, 2021.

Anton Kolotilin. Optimal information disclosure: A linear programming approach. *Theoretical Economics*, 13(2):607–635, 2018.

Daniel Krähmer. Information design and strategic communication. *American Economic Review: Insights*, 3(1):51–66, 2021.

Sophie Kreutzkamp and Yichuan Lou. Persuasion without ex-post commitment. *Journal of Economic Theory*, page 106058, 2025.

Yichuan Lou. Optimal delegation: Coarsification through information design. 2023.

Qianjun Lyu and Wing Suen. Information design in cheap talk. 2024.

Mark Machina and John Pratt. Increasing risk: some direct constructions. *Journal of Risk and Uncertainty*, 14:103–127, 1997.

Jeffrey Mensch. Monotone persuasion. *Games and Economic Behavior*, 130:521–542, 2021.

Luis Rayo and Ilya Segal. Optimal information disclosure. *Journal of political Economy*, 118(5): 949–987, 2010.

Michael Rothschild and Joseph Stiglitz. Increasing risk: I. a definition. *Journal of Economic Theory*, 2(3):225–243, 1970.

Denis Shishkin. Evidence acquisition and voluntary disclosure. 2024.

Joel Watson. Information transmission when the informed party is confused. *Games and Economic Behavior*, 12(1):143–161, 1996.

A Appendix

Proof of Proposition 1

We are interested in the area below c_π, which is characterized by the intersection points of segments of c_π. It is clear that Segment i and Segment $(i+1)$ intersect at x_{i+1}, the $(i+1)$-th posterior mean. Let y_{i+1} be the corresponding vertical coordinate of the intersection point. Substitute x_{i+1} back to either Segment i or Segment $(i+1)$, we get that

$$y_{i+1} = \sum_{j=1}^{i} w_j(x_{i+1} - x_j). \tag{10}$$

In general, the area below c_π can be decomposed naturally into a triangle and several trapezoids, by vertically slicing at each x_i. An example with $N = 3$ is showing in Figure 5. Working towards a general conclusion, let's first consider the area excluding the last trapezoid (i.e., the trapezoid with the edge $(x_N, 0)$ – $(1, 0)$), denoting it by S_N.

When $N = 2$, the area excluding the last trapezoid is just the triangle, with the area

$$S_2 \equiv \frac{1}{2}(x_2 - x_1)y_2 = \frac{1}{2}(x_2 - x_1)w_1(x_2 - x_1) = \frac{1}{2}w_1(x_2 - x_1)^2.$$

For $N = 3$,

$$\begin{aligned} S_3 &\equiv \frac{1}{2}(x_2 - x_1)y_2 + \frac{1}{2}(x_3 - x_2)(y_2 + y_3) \\ &= \frac{1}{2}w_1(x_2 - x_1)^2 + \frac{1}{2}(x_3 - x_2)\left[w_1(x_2 - x_1) + w_1(x_3 - x_1) + w_2(x_3 - x_2)\right] \\ &= \frac{1}{2}\left[w_1(x_3 - x_1)^2 + w_2(x_3 - x_2)^2\right]. \end{aligned}$$

This naturally leads to the following: For any finite N,

$$S_N = \frac{1}{2}\sum_{j=1}^{N-1} w_j(x_N - x_j)^2.$$

We can show this by induction. This has been verified for $N = 2$. Suppose that this holds for $N = k$,

$$S_k = \frac{1}{2}\sum_{j=1}^{k-1} w_j (x_k - x_j)^2.$$

Consider $N = k + 1$. Observe that

$$\begin{aligned} S_{k+1} &= S_k + \frac{1}{2}(x_{k+1} - x_k)(y_k + y_{k+1}) \\ &= \frac{1}{2}\sum_{j=1}^{k-1} w_j (x_k - x_j)^2 + \frac{1}{2}(x_{k+1} - x_k)\left[\sum_{j=1}^{k-1} w_j (x_k - x_j) + \sum_{j=1}^{k} w_j (x_{k+1} - x_j)\right]. \end{aligned} \tag{11}$$

The first line follows from the fact that S_{k+1} can be decomposed into the last trapezoid it includes (i.e., the trapezoid with the edge $(x_k, 0)$ – $(x_{k+1}, 0)$) and what left. Yet what left is the same "shape" as the shapes included in S_N. The second line follows from the induction assumption and (10).

For any $i = 1, \cdots, k-1$, let's collect the terms in (11) that contains the coefficient w_i. For now, let's also omit the coefficient $1/2$ for the ease of notations. These terms are

$$w_i (x_k - x_i)^2 + w_i (x_{k+1} - x_k)\left[(x_k - x_i) + (x_{k+1-x_i})\right] = w_i (x_{k+1} - x_i)^2.$$

What left is just the term contains the coefficient w_k, which is just $w_k (x_{k+1} - x_k)^2$. Substitute these back to (11) and put the coefficient $1/2$ back,

$$S_{k+1} = \frac{1}{2}\sum_{j=1}^{k} w_j (x_{k+1} - x_j)^2,$$

which is exactly what we need.

Now let's consider the last trapezoid. The area, denoted by T_N, is

$$T_N \equiv \frac{1}{2}\left[y_N + c_\pi(1)\right](1 - x_N),$$

where

$$\begin{aligned} c_\pi(1) = c_{\overline{\pi}}(1) &= \int_0^1 F(t)\mathrm{d}t \\ &= F(t)t|_0^1 - \int_0^1 t f(t)\mathrm{d}t \\ &= 1 - \sum_{j=1}^{N} w_j x_j. \end{aligned}$$

Following the expressions above,

$$2(S_N + T_N) = \sum_{j=1}^{N-1} w_j (x_N - x_j)^2 + (1 - x_N)\left[\sum_{j=1}^{N-1} w_j (x_N - x_j) + 1 - \sum_{j=1}^{N} w_j x_j\right]. \tag{12}$$

Notice that we consider $2(S_N + T_N)$ to remove the coefficient $1/2$ from the right hand side. Collect terms that contains the coefficient w_i, for $i = 1, \cdots, N-1$,

$$w_i(x_N - x_i)^2 + w_i\left[(x_N - x_i) - x_i\right](1 - x_N) = w_i x_N - 2w_i x_i + w_i x_i^2.$$

This shows, by substituting this back to (12),

$$\begin{aligned}
2(S_N + T_N) &= \sum_{j=1}^{N-1}\left(w_j x_N - 2w_j x_j + w_j x_j^2\right) + (1 - w_N x_N)(1 - x_N) \\
&= 1 + \left(\sum_{j=1}^{N-1} w_j\right) x_N - x_N - w_N x_N - 2\sum_{j=1}^{N-1} w_j x_j + w_N x_N^2 + \sum_{j=1}^{N-1} w_j x_j^2 \\
&= 1 - 2\sum_{j=1}^{N} w_j x_j + \sum_{j=1}^{N} w_j x_j^2.
\end{aligned}$$

Therefore,

$$S_N + T_N = \frac{1}{2} EU^R(\pi) + \frac{1}{2} - \mu,$$

where μ is the prior mean. This concludes the proof. ■

Proof of Proposition 5

We start by noting that the conclusion in Gentzkow and Kamenica (2016) is not restricted to the case that the prior distribution has the support $[0, 1]$. Instead, it works on all closed intervals. The details are as follows:

- The part showing that the integrated CDF of an information structure must be between $c_{\overline{\pi}}$ and $c_{\underline{\pi}}$ uses the property of *mean-preserving spread*, which is not a concept restricted on $[0, 1]$.
- The part showing that a convex function between $c_{\overline{\pi}}$ and $c_{\underline{\pi}}$ corresponds to an information structure uses *Proposition 1* from Kolotilin (2014), which is shown under a more general interval $[\underline{r}, \overline{r}]$.

In particular, we will later use Gentzkow and Kamenica (2016) on $[0, m]$ for some $m < 1$.

Let c_π be the integrated CDF for some finite π such that segment i of c_π is tangent to $c_{\overline{\pi}}$ at some horizontal coordinate $m \in (0, 1)$. This implies the tangency point is

$$\left(m, \int_0^m F(t)\mathrm{d}t\right).$$

Segment i is a straight line passes through this point, with the slope of $F(m)$, the slope of $c_{\overline{\pi}}$ at this point. Therefore, we can write Segment i as

$$c_\pi(x) = F(m)x + c$$

for some constant c. Substitute the coordinate of the tangency point,

$$\begin{aligned} c &= \int_0^m F(t)\mathrm{d}t - mF(m) \\ &= tF(t)|_0^m - \int_0^m t\mathrm{d}F(t) - mF(m) \\ &= -\int_0^m t\mathrm{d}F(t) \\ &= -F(m)\mathbb{E}(\theta \mid \theta \in [0,m]). \end{aligned}$$

The second line uses integration by parts, the third line uses the fact that $F(0) = 0$, and the last line uses the definition of conditional expectation of θ on $[0, m]$. Therefore, the expression of Segment i is

$$c_\pi(x) = F(m)x - F(m)\mathbb{E}(\theta \mid \theta \in [0,m]). \tag{13}$$

Let's now take a detour and consider an information structure on $[0, m]$. Let the prior distribution on $[0, m]$ be

$$\tilde{F}(x) = \frac{F(x)}{F(m)}.$$

That is, we consider the original prior distribution F conditioned on $[0, m]$. An information structure based on $\tilde{F}$ has the upper bound

$$\tilde{c}_{\overline{\pi}}(x) = \int_0^x \tilde{F}(t)\mathrm{d}t = \frac{1}{F(m)} c_{\overline{\pi}}(x).$$

The lower bound $\tilde{c}_{\underline{\pi}}$ is a piecewise linear function which is 0 before $\mathbb{E}(\theta \mid \theta \in [0,m])$, and

$$x - \mathbb{E}(\theta \mid \theta \in [0,m])$$

for $x \in [\mathbb{E}(\theta \mid \theta \in [0,m]), m]$. Notice that $\tilde{c}_{\overline{\pi}}$ and $\tilde{c}_{\underline{\pi}}$ coincide at m.

Now consider c_π again, but scaled by $1/F(m)$. After this constant scaling, the new Segment 0 is still on the horizontal axis, and the new Segment i is, by using (13), exactly $\tilde{c}_{\underline{\pi}}$. Considering c_π only on $[0, m]$, the scaled version

$$\frac{c_\pi}{F(m)}$$

is a $(i+1)$-segment piecewise linear convex function between $\tilde{c}_{\underline{\pi}}$ and $\tilde{c}_{\overline{\pi}}$. This implies the scaled version, $c_\pi/F(m)$, defines an information structure on $[0, m]$. Correspondingly, c_π then also defines an information structure such that no signal realization sent at the states below m is also sent above m. (To further confirm this, one can check that the weighted average of x_i before m, using the scaled $c_\pi/F(m)$, is $\mathbb{E}(\theta \mid \theta \in [0,m])$.)

Conversely, consider an information structure such that, for some $m \in (0, 1)$, no signal realization sent at the states below m is also sent above m. One can imagine that there is a "straight line"

that divide $[0,1]$ into two independent parts, $[0,m]$ and $(m,1]$. Suppose that there are i posterior means induced before m; that is,

$$0 < x_1 < \cdots < x_i < m.$$

(Immediately, it is impossible that $x_i = m$ for some x_i. Otherwise, given that we have ruled out posterior means with zero weights, the signal realization that induces x_i must be sent above and below x_i.)

What is the expression of Segment i? Recall that in general, Segment i has the expression

$$c_\pi(x) = \left(\sum_{j=1}^{i} w_j\right) x - \sum_{j=1}^{i} w_j x_j.$$

Because of the straight line formulation,

$$\sum_{j=1}^{i} w_j = F(m),$$

since the signal realizations that induce $x_1, \cdots, x_i$ are sent on and only on $[0,m]$, and no other signal realizations are sent on $[0,m]$. For the same reason, $\sum_{j=1}^{i} w_j x_j$ must be the conditional mean on $[0,m]$ scaled by the weight, namely,

$$\sum_{j=1}^{i} w_j x_j = F(m)\mathbb{E}(\theta \mid \theta \in [0,m]).$$

But this means the expression of Segment i is exactly (13), which then shows that Segment i is tangent to $c_{\overline{\pi}}$. ■

Proof of Corollary 8

It suffices to show the following lemma.

Lemma 17. *Suppose x_i and x_{i+1} are induced and only induced on states $[\ell, h]$. Let the weight of s_i (s_{i+1}) being sent be w_i (w_{i+1}). Regardless of the original information structure, the following modification leads to a payoff equivalent structure:*

- *In states $[\tilde{\ell}, \tilde{h}]$, signal s_{i+1} is sent, such that $\ell < \tilde{\ell} < \tilde{h} \leqslant h$, and*

$$\int_{\tilde{\ell}}^{\tilde{h}} f(\theta)\mathrm{d}\theta = w_{i+1}, \quad \text{and} \quad \frac{1}{w_{i+1}} \int_{\tilde{\ell}}^{\tilde{h}} \theta f(\theta)\mathrm{d}\theta = x_{i+1}.$$

- *In states $[\ell, \tilde{\ell}) \cup (\tilde{h}, h]$, signal s_i is sent.*

We first consider the existences of $\tilde{\ell}$ and $\tilde{h}$. Let $\tilde{h}(t)$ be an implicit function defined by

$$\int_t^{\tilde{h}(t)} f(\theta)\mathrm{d}\theta = w_{i+1}.$$

Since f is assumed to have full support, $\tilde{h}(t)$ is strictly increasing. It is also clear that $\tilde{h}(\cdot)$ is continuous. To show this, first recall that f is continuous on $[0,1]$, so that there exists m and M such that $m \leqslant f \leqslant M$ for all $\theta \in [0,1]$. Let $\varepsilon > 0$ be given. We show that for any t_1, for t_2 satisfying $|t_2 - t_1| < \delta$, $|\tilde{h}(t_2) - \tilde{h}(t_1)| < \varepsilon$, where $\delta = \varepsilon m/M$. For simplicity, we show only when $t_2 > t_1$, and the other side is similar.

By definition, $\int_{t_1}^{\tilde{h}(t_1)} f(\theta)\mathrm{d}\theta = \int_{t_2}^{\tilde{h}(t_2)} f(\theta)\mathrm{d}\theta$, or equivalently,

$$\int_{t_1}^{t_2} f(\theta)\mathrm{d}\theta = \int_{\tilde{h}(t_1)}^{\tilde{h}(t_2)} f(\theta)\mathrm{d}\theta.$$

The left hand side is less than δM. Substitute $\delta = \varepsilon m/M$, this means

$$\int_{\tilde{h}(t_1)}^{\tilde{h}(t_2)} f(\theta)\mathrm{d}\theta < \varepsilon m.$$

Since $f \geqslant m$, $\tilde{h}(t_2) - \tilde{h}(t_1) < \varepsilon$.

Define function

$$\tilde{x}(t) = \frac{1}{w_{i+1}} \int_t^{\tilde{h}(t)} \theta f(\theta)\mathrm{d}\theta,$$

which is the posterior mean induced by the cell $[t, \tilde{h}(t)]$. Notice that this is also continuous (which can be shown similarly and the details are omitted). Let $\bar{t}$ be the point in $[\ell, h]$ such that $\tilde{h}(\bar{t}) = h$. Since $\tilde{x}(\ell) < x_{i+1} < \tilde{x}(\bar{t})$, by the intermediate value theorem, there exists $\tilde{\ell}$ such that $x_{\tilde{\ell}} = x_{i+1}$. In turn, this is exactly the ℓ we are looking for. Also, $\tilde{h} = \tilde{h}(\ell)$.

The rest of the proof is clear: since the structure after modifications keeps the same x_{i+1}, w_{i+1}, it must keep the same w_i, which then shows x_i must be unchanged as well. Given (2), this is all we need. ■

Proof of Theorem 9

The Receiver's ex-ante expected payoff under a four-signal structure is

$$\sum_{i=1}^{4} w_i x_i^2.$$

Substitute $x_2 = x_1 + 2b$, $x_3 = x_4 - 2b$, $w_1 = 2x_1$, $w_4 = 2(1 - 2x_4)$, and w_2 and w_3 from (4), the payoff is half of the following (That is, for simplicity, we drop the coefficient $1/2$ in the payoff in the remaining part of the proof.)

$$x_1 + x_4 - 2x_1x_4 - 8b^2(1 + 2x_1 - 2x_4) - 4b(x_1 + 2x_1^2 - 4x_1x_4 + x_4(2x_4 - 1)). \tag{14}$$

The Receiver chooses x_1 and x_4 to maximize this subject to $0 < x_1 < x_4 < 1$ and

$$x_4 - x_1 \geqslant 6b.$$

This is because the Receiver must also fit x_2 and x_3 in between, so that the IC condition implies there must be at least $6b$ distance between x_1 and x_4. The Hessian matrix of the objective (i.e., payoff) function in (14),

$$H = \begin{bmatrix} -16b & 16b-2 \\ 16b-2 & -16b \end{bmatrix}$$

Given the range of b, H is negative definite, showing that (14) is strictly concave in both variables. This implies that we can use the Kuhn-Tucker conditions to solve the optimzer. The Lagrange function is

$$\begin{aligned}\mathcal{L}(x_1, x_4, \lambda) = {} & x_1 + x_4 - 2x_1x_4 - 8b^2(1 + 2x_1 - 2x_4) - 4b(x_1 + 2x_1^2 - 4x_1x_4 + x_4(2x_4 - 1)) \\ & + \lambda(x_4 - x_1 - 6b).\end{aligned}$$

The Kuhn-Tucker conditions are

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial x_1} &= 1 - 16b^2 - 4b(1 + 4x_1 - 4x_4) - 2x_4 - \lambda = 0, \\ \frac{\partial \mathcal{L}}{\partial x_4} &= 1 + 16b^2 + 4b(1 + 4x_1 - 4x_4) - 2x_1 + \lambda = 0, \\ \lambda \cdot \frac{\partial \mathcal{L}}{\partial \lambda} &= \lambda(x_4 - x_1 - 6b) = 0, \qquad \lambda \geqslant 0.\end{aligned}$$

Notice that the conditions are slightly simplified since we know that x_1 and x_4 must be strictly between 0 and 1. We also know that the solution must exist since the objective function is strictly concave and the feasible region is convex. The Kuhn-Tucker conditions give

$$x_1 = \frac{1}{2} - 3b, \quad x_4 = \frac{1}{2} + 3b, \quad \text{and} \quad \lambda = 10b(8b - 1).$$

Notice that since $\lambda > 0$, the constraint must bind, which implies that the IC condition binds everywhere. Therefore,

$$\begin{aligned}x_2 = \frac{1}{2} - b, \quad x_3 = \frac{1}{2} + b, \\ w_2 = w_3 = 6b - \frac{1}{2}.\end{aligned}$$

Notice that when $b = 1/8$, the distribution of posterior mean here is exactly the one induced by the four-cell evenly partitional structure.

We then need to verify that such distribution of posterior means can be induced by an information structure. Of course, one can check that the corresponding integrated CDF lies below

$c_{\overline{\pi}}(x) = x^2/2$. Alternatively, since all mixing regions only induces two posterior means, a mixing region $[\ell, h]$ that induces x_i and x_{i+1} can be generated by an information structure if and only if

$$x_i + (x_i - \ell) \geqslant x_{i+1} - (h - x_{i+1}),$$

and the equality holds if the mixing is degenerate; that is, x_i and x_{i+1} are induced by two partition cells. The idea is as follows. Imagine x_i and x_{i+1} were induced by partition cells. If the right boundary of the cell induces x_i was smaller than the left boundary of the cell induces x_{i+1}, the two posterior would be two spreaded from each other such that evenly partitional structures could not generate such posterior means. In turn, the two cells must overlap with each other in a mixing structure. It turns out that this is also all one need for feasibility. It can be verified that the structure above is always feasible.

Given the Receiver's payoff is $\sum_{i=1}^{N} w_i x_i$, the payoffs of the structure propsed in Theorem 9 is

$$-96b^3 + 17b^2 + \frac{1}{4}. \tag{15}$$

The payoff of the three-cell even-partition strucuture is 35/108. Notice that in $(1/8, 1/6)$, (15) is strictly monotone decreasing. When $b = 1/8$, (15) is 21/64, greater than 35/108. When $b = 1/6$, the structure proposed in Theorem 9 reduces to a two-signal structure with payoff 5/18, lower than 35/108. The existence of $\bar{b}$ is then guaranteed by the intermediate value theorem, and it must be unique. To solve for $\bar{b}$, let

$$-96b^3 + 17b^2 + \frac{1}{4} = \frac{35}{108},$$

which gives $\bar{b} \approx 0.134309$. ■

Proof of Theorem 10

It suffices to verify that Lemma 2, Lemma 3, and Lemma 4 continue to hold. For Lemma 2, we consider the case that x_1 is not generated from a partition cell, and the discussion regarding x_N is similar. Consider the local perturbation that replaces x_1 with $x_1 - \delta$ for some small $\delta > 0$. Since the original information structure is IC,

$$x_2 - x_1 \geqslant 2b(x_1).$$

Now consider $x_2 - (x_1 - \delta) - 2b(x_1 - \delta)$, which is

$$x_2 - x_1 + \delta - 2[\alpha + \rho(x_1 - \delta)] = x_2 - x_1 - 2b(x_1) + (1 + 2\rho)\delta. \tag{16}$$

Since $\rho \geqslant -1/2$, this is positive, showing that IC is preserved.

For Lemma 3, notice that the local perturbation does not change the value of x_i. Under Assumption 1, this means the bias value $\beta_i = b(x_i)$ are also not changed, so that the IC constraints are preserved.

For Lemma 4, first consider the perturbation in the right panel of Figure 4, where we increase x_{i+1} by δ. As mentioned above, the perturbation affects two IC constraints. Notice that $x_{i+1} - x_i \geqslant 2b(x_i)$. Since the location of x_i is unchanged, and x_{i+1} increases, this IC still holds after perturbation. For $x_{i+2} - x_{i+1} > 2b(x_{i+1})$, since the change of x_{i+1} is δ, the change of $b(x_{i+1})$ is $\rho\delta$, and the IC constraint is relaxed, the IC constraint is preserved when δ is small enough.

For the other direction, suppose that $x_i - x_{i-1} > 2b(x_{i-1})$, and we want to decrease x_i by δ. Similarly, it suffices to consider the IC constraint $x_{i+1} - x_i \geqslant 2b(x_i)$. Here we need to evaluate

$$x_{i+1} - (x_i - \delta) - 2b(x_i - \delta).$$

This is nonnegative, and the argument is similar to (16), which indicates the IC constraint is preserved.

Proof of Proposition 11

Suppose not. That is, suppose that there is a signal s_j that are sent at states above and below m (with positive weights on both sides). Let G be the posterior mean distribution. We first calculate $c_{\overline{\pi}}(m)$ and $c_\pi(m)$.

By definition and integration by parts,

$$\begin{aligned} c_{\overline{\pi}}(m) &= \int_0^m F(\theta)\mathrm{d}\theta = \theta F(\theta)|_0^m - \int_0^m \theta \mathrm{d}F(\theta) \\ &= \int_0^m (m-\theta)\mathrm{d}F(\theta) \\ &= \int_0^1 \max\{m-\theta, 0\}\mathrm{d}F(\theta). \end{aligned}$$

Similarly,

$$c_\pi(m) = \int_0^1 \max\{m-x, 0\}\mathrm{d}G(x).$$

Now consider signal s_j. Since $\phi(z) \equiv \max\{m-z, 0\}$ is a convex function, Jensen inequality implies

$$\underbrace{\phi(\mathbb{E}[\theta \mid s_j])}_{\phi(x_j)} < \mathbb{E}[\phi(\theta) \mid s_j].$$

The inequality is strict since s_j is sent both above and below m (i.e., on both arms of ϕ function). For all other signals, the same Jensen inequality implies $\phi(\mathbb{E}[\theta \mid s_i]) \leqslant \mathbb{E}[\phi(\theta) \mid s_i]$. Taking expectations on both sides, we now have

$$\int_0^1 \max\{m-x, 0\}\mathrm{d}G(x) < \int_0^1 \max\{m-\theta, 0\}\mathrm{d}F(\theta),$$

where on the right hand side we use law of iterated expectations. The inequality remains strict since s_j is sent with positive weight. But this simply means $c_\pi(m) < c_{\overline{\pi}}(m)$, which contradicts the assumption that the two functions has a contact point at m.

Proof of Theorem 12

First, consider an interior mass point $m \in (0,1)$ which itself may or may not be an induced posterior mean. We consider the left-side condition that $m - x_i \geqslant 2b$ but $x_i - x_{i-1} > 2b$, and the right-side condition is similar. Consider a clockwise rotation of the part c_π between x_i and m while keeping $c_\pi(m)$ fixed. For small rotations, c_π is still a convex function. Since $F(m^-) < c'_\pi(m^-)$, c_π is still below $c_{\overline{\pi}}$, so that c_π is still feasible. This rotation creates or enlarges the kink of c_π at m, which suggests that the mass point m is revealed with (a larger) positive probability according to Proposition 11. Therefore, m itself is now an induced posterior mean after the modification with a higher probability. Since $m - x_i \geqslant 2b$ and $x_i - x_{i-1} > 2b$, the modification preserves IC constraint. Since the rotation increases the area below c_π, this modification strictly increases the Receiver's payoff. This is pictured in Figure 11.

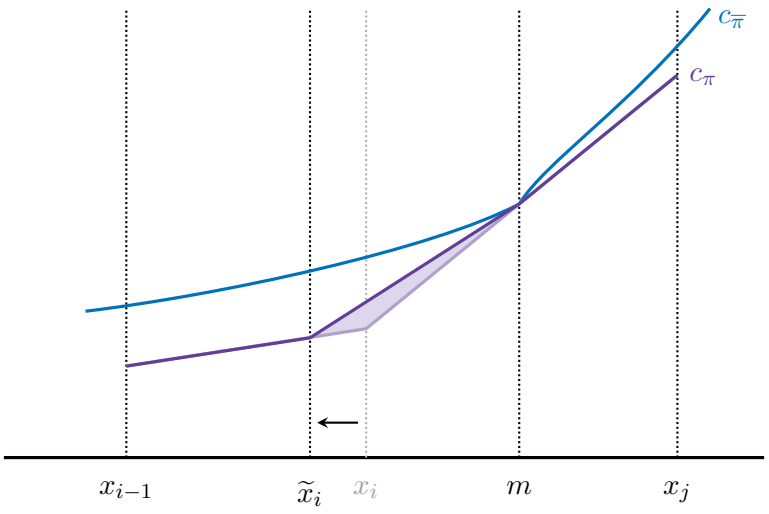


Figure 11 Profitable local adjustments with an interior mass point

Next, consider boundary mass points. We consider the case where there is a mass point at state 0. The arguments for a mass point at state 1 are symmetric.

- Suppose first that $0 < r_0 < p_0$ but $x_2 - x_1 > 2b$. The argument is similar to the one above regarding interior mass points. The Receiver can make a small counterclosewise rotation at the first segment of c_π, increase the area below c_π, while maintaining the feasibility and IC constraints. This is shown in the left panel of Figure 12.
- Suppose next that $0 < r_0 \leqslant p_0$ and $c_\pi < c_{\overline{\pi}}$ for all $x \in [x_1, x_2]$, but $x_1 > 2b$. Then we can apply the adjustment in Lemma 2 by rotating the second segment of c_π clockwise. This is shown in the right panel of Figure 12. Essentially, the only reason not to generate x_1 from a partition cell is the possibility of fully revealing state 0. But if the IC constraint between x_1 and 0 is not binding, then the designer can make the signal more accurate for x_1 and x_2.

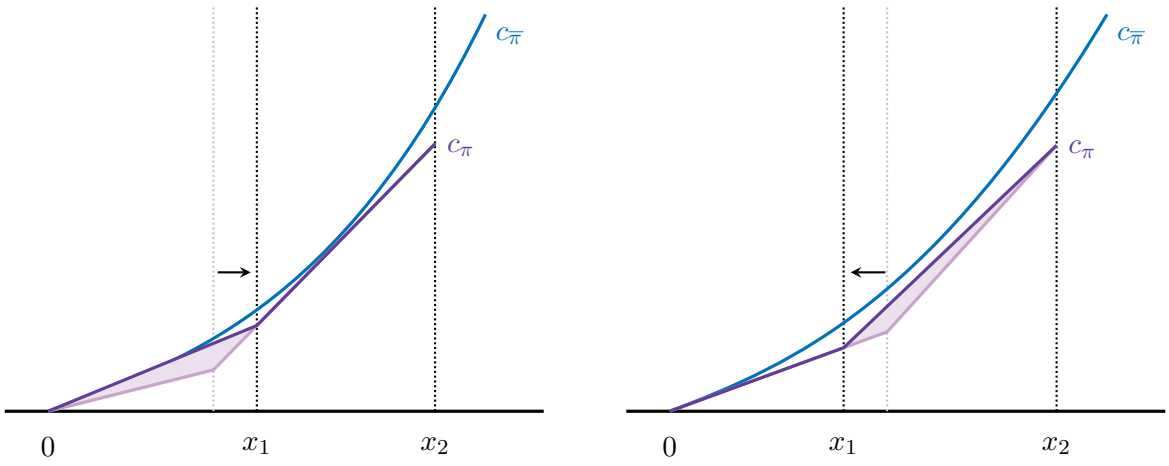


Figure 12 Profitable local adjustments when state 0 is a mass point

Proof of Proposition 15

To see that (8) must bind, suppose the contrary. That is, if $\pi^\star$ is the optimal structure, $E - E(\pi^\star) = \varepsilon > 0$. Pick an arbitrary signal realization s_k (that induces x_k in $\pi^\star$). Since $g(\cdot)$ is continuous, there exists $\bar{z} > 0$ such that

$$0 < g(\bar{z}) - (-w_k \ln w_k) < \frac{\varepsilon}{2}.$$

We now proposed the following modified information structure π'. Pick some $m \in (0,1)$ (defined later), and create a new signal s'_k. Then,

$$\pi'(s \mid \omega) = \begin{cases} \pi^\star(s \mid \omega) & \text{if } s \neq s_k \text{ or } s'_k, \\ \pi(s_k \mid \omega) & \text{if } s = s'_k, \\ 0 & \text{if } s = s_k \end{cases} \qquad \text{if } \omega < m,$$

and

$$\pi'(s \mid \omega) = \begin{cases} \pi^\star(s \mid \omega) & \text{if } s \neq s'_k, \\ 0 & \text{if } s = s'_k \end{cases} \qquad \text{if } \omega \geqslant m.$$

That is, we split the signal realization s_k in two by replace s_k by s'_k for all states below m. m is defined such that the overall weight of sending s'_k is less than $\bar{z}$, i.e.,

$$\int_0^m \pi^\star(s_k \mid \omega) f(\omega) \mathrm{d}\omega < \bar{z},$$

which is always feasible since f is continuous and every signal realization is assumed to be generated with positive weights. This modification only increase the entropy less than $\varepsilon/2$ so that (8) is still satisfied. Yet it is clear that the designer is better off. ■

Proof of Proposition 16

Consider a k-cell structure for some arbitrary $k \geqslant 3$. (The problem is more or less trivial if there are at most two cells.) For a given upper bound E on the entropy, the information designer's problem is

$$\begin{aligned} \min_{w_1,\cdots,w_{k-1}} \quad & \sum_{i=1}^{k-1} w_i^3 + \left(1 - \sum_{i=1}^{k-1} w_i\right)^3 \\ \text{subject to} \quad & -\sum_{i=1}^{k-1} w_i \ln w_i - \left(1 - \sum_{i=1}^{k-1} w_i\right) \ln \left(1 - \sum_{i=1}^{k-1} w_i\right) \leqslant E, \end{aligned}$$

and $w_i \geqslant 0$ for all $i = 1, \cdots, k-1$. The Lagrangian is

$$\mathcal{L} = \sum_{i=1}^{k-1} w_i^3 + \left(1 - \sum_{i=1}^{k-1} w_i\right)^3 + \lambda \left[-E - \sum_{i=1}^{k-1} w_i \ln w_i - \left(1 - \sum_{i=1}^{k-1} w_i\right) \ln \left(1 - \sum_{i=1}^{k-1} w_i\right)\right].$$

For a generic i, the FOC is

$$\frac{\partial \mathcal{L}}{\partial w_i} = 3w_i^2 - 3\left(1 - \sum_{i=1}^{k-1} w_i\right)^2 + \lambda \left[\ln \left(1 - \sum_{i=1}^{k-1} w_i\right) - \ln w_i\right] = 0. \tag{17}$$

Let $1 - w_1 - \cdots - w_{k-1} = w_k$. The FOC indicates that, if $w_i \neq w_k$ for all $i = 1, \cdots, k-1$,

$$\lambda = \frac{3(w_i^2 - w_k^2)}{\ln w_i - \ln w_k}.$$

This further implies that for $i \neq j$,

$$\frac{(w_i/w_k)^2 - 1}{(w_j/w_k)^2 - 1} = \frac{\ln(w_i/w_k)}{\ln(w_j/w_k)}.$$

Since the function

$$g(x) = \frac{x^2 - 1}{\ln x} \tag{18}$$

is strictly increasing, $w_i/w_k = w_j/w_k$, or simply, $w_i = w_j$. This shows that in an optimal structure, there are at most two cell sizes, and all but one cell share the same size.

Let's further check the second order conditions. Using (17),

$$\begin{aligned} \frac{\partial^2 \mathcal{L}}{\partial w_i^2} &= 6w_i + 6w_k - \frac{\lambda}{w_i} - \frac{\lambda}{w_k}, \\ \frac{\partial^2 \mathcal{L}}{\partial w_i \partial w_j} &= 6w_k - \frac{\lambda}{w_k}. \end{aligned}$$

We can then write the bordered Hessian matrix H_k^B. Since $w_1 = \cdots = w_{k-1}$ in the solution of the FOCs, we will write all w_i as w_1 for $i = 1, \cdots, k-1$. Then,

$$H_k^B = \begin{bmatrix} 0 & \ln\frac{w_k}{w_1} & \ln\frac{w_k}{w_1} & \cdots & \ln\frac{w_k}{w_1} \\ \ln\frac{w_k}{w_1} & 6w_1 + 6w_k - \frac{\lambda}{w_1} - \frac{\lambda}{w_k} & 6w_k - \frac{\lambda}{w_k} & \cdots & 6w_k - \frac{\lambda}{w_k} \\ \ln\frac{w_k}{w_1} & 6w_k - \frac{\lambda}{w_k} & 6w_1 + 6w_k - \frac{\lambda}{w_1} - \frac{\lambda}{w_k} & \cdots & 6w_k - \frac{\lambda}{w_k} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \ln\frac{w_k}{w_1} & 6w_k - \frac{\lambda}{w_k} & 6w_k - \frac{\lambda}{w_k} & \cdots & 6w_1 + 6w_k - \frac{\lambda}{w_1} - \frac{\lambda}{w_k} \end{bmatrix}$$

To calculate the determinant of this matrix, we first subtract Row 2 from Rows $3, 4 \cdots, k$. We then subtract Column k from Columns $2, 3 \cdots k-1$. All these operations do not change the determinant. The matrix has been transformed into the following matrix,

$$\begin{bmatrix} 0 & 0 & 0 & 0 & \cdots & 0 & \ln\frac{w_k}{w_1} \\ \ln\frac{w_k}{w_1} & 6w_1 - \frac{\lambda}{w_1} & 0 & 0 & \cdots & 0 & 6w_k - \frac{\lambda}{w_k} \\ 0 & -6w_1 + \frac{\lambda}{w_1} & 6w_1 - \frac{\lambda}{w_1} & 0 & \cdots & 0 & 0 \\ 0 & -6w_1 + \frac{\lambda}{w_1} & 0 & 6w_1 - \frac{\lambda}{w_1} & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & -6w_1 + \frac{\lambda}{w_1} & 0 & 0 & \cdots & 6w_1 - \frac{\lambda}{w_1} & 0 \\ 0 & -12w_1 + \frac{2\lambda}{w_1} & -6w_1 + \frac{\lambda}{w_1} & -6w_1 + \frac{\lambda}{w_1} & \cdots & -6w_1 + \frac{\lambda}{w_1} & 6w_1 - \frac{\lambda}{w_1} \end{bmatrix}$$

We can then expand this matrix from the first column. It suffices to consider the determinant of the following matrix

$$\begin{bmatrix} 0 & 0 & 0 & \cdots & 0 & \ln\frac{w_k}{w_1} \\ -6w_1 + \frac{\lambda}{w_1} & 6w_1 - \frac{\lambda}{w_1} & 0 & \cdots & 0 & 0 \\ -6w_1 + \frac{\lambda}{w_1} & 0 & 6w_1 - \frac{\lambda}{w_1} & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ -6w_1 + \frac{\lambda}{w_1} & 0 & 0 & \cdots & 6w_1 - \frac{\lambda}{w_1} & 0 \\ -12w_1 + \frac{2\lambda}{w_1} & -6w_1 + \frac{\lambda}{w_1} & -6w_1 + \frac{\lambda}{w_1} & \cdots & -6w_1 + \frac{\lambda}{w_1} & 6w_1 - \frac{\lambda}{w_1} \end{bmatrix} \tag{19}$$

Add Column $2, \cdots, k-2$ to the first column, and then switch Column 1 and Column $k-1$ to get

$$\begin{bmatrix} \ln\frac{w_k}{w_1} & 0 & 0 & \cdots & 0 & 0 \\ 0 & 6w_1 - \frac{\lambda}{w_1} & 0 & \cdots & 0 & 0 \\ 0 & 0 & 6w_1 - \frac{\lambda}{w_1} & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 6w_1 - \frac{\lambda}{w_1} & 0 \\ 6w_1 - \frac{\lambda}{w_1} & -6w_1 + \frac{\lambda}{w_1} & -6w_1 + \frac{\lambda}{w_1} & \cdots & -6w_1 + \frac{\lambda}{w_1} & (k-1)\left(-6w_1 + \frac{\lambda}{w_1}\right) \end{bmatrix}$$

This is a triangular matrix, and the determinant is the product of the diagonal elements,

$$-(k-1)\ln\frac{w_k}{w_1}\left(6w_1-\frac{\lambda}{w_1}\right)^{k-2}.$$

Also, as we have switch columns once, the determinant of (19) needs to be multiplied by an additional (-1). The overall determinant of H_k^B is then

$$-(k-1)\left(\ln\frac{w_k}{w_1}\right)^2\left(6w_1-\frac{\lambda}{w_1}\right)^{k-2}. \tag{20}$$

We should note that all designer minor of H_k^B share the same structure, since they are just H_{k-1}^B, H_{k-2}^B, etc. Therefore, if the determinant H_k^B and the determinant of all designer minors of H_k^B are negative as required by the second order conditions, it must be the case that

$$6w_1-\frac{\lambda}{w_1}>0,$$

so that (20) is negative for all k. We already know that $\lambda=3(w_1^2-w_k^2)/\ln(w_1/w_k)$, so that we need to consider the sign of

$$6w_1-\frac{3(w_1^2-w_k^2)}{w_1\ln\frac{w_1}{w_k}}=3w_1\left[2+\frac{1-\left(\frac{w_k}{w_1}\right)^2}{\ln\frac{w_k}{w_1}}\right].$$

Notice that the terms in the bracket is $2-g(w_k/w_1)$. Since $\lim_{x\to1}g(x)=2$, and $g(\cdots)$ is strictly increasing, we know that $6w_1-(\lambda/w_1)>0$ if and only if $w_k/w_1<1$, or $w_k<w_1$. Therefore, any solution with $w_k>w_1$ will violate the second order conditions, implying that the solution does not attain a local minimum. That is, the optimal structure must have the following features:

- It is either an evenly partitional structure; or
- It is a partitional structure with two types of cell sizes. All but one cell share the same, larger cell size, and the one cell left has the smaller cell size.

Let's now return to the case where the entropy upper bound E is $-\ln(K+\varepsilon)$ for some $\varepsilon\in[0,1)$. Consider a $(K+2)$-cell structure with $w_{K+2}=\delta$, so that $w_1=\cdots=w_{K+1}=(1-\delta)/(K+1)$. The entropy of the structure is

$$-\left[(K+1)\frac{1-\delta}{K+1}\ln\frac{1-\delta}{K+1}+\delta\ln\delta\right]=-(1-\delta)\ln\frac{1-\delta}{K+1}-\delta\ln\delta.$$

If $\delta<(1-\delta)/(K+1)$,

$$-(1-\delta)\ln\frac{1-\delta}{K+1}-\delta\ln\delta>-(1-\delta)\ln\frac{1-\delta}{K+1}-\delta\ln\frac{1-\delta}{K+1}=\ln\frac{K+1}{1-\delta}>\ln(K+1).$$

More generally, for any generic k, the structure such that $k-1$ cells share a larger cell size, and the remaining one cell has a smaller cell size must have an entropy greater than $\ln k$. In other

words, for the upper bound $-\ln(K+\varepsilon)$ for some $\varepsilon \in [0,1)$, a structure with $K+2$ or more cells that satisfies the first and the second order conditions above must exceed the upper bound of the entropy. Thus, the optimal structure is at most $K+1$ cell. When $\varepsilon = 0$, the $(K+1)$-cell structure is also impossible (since it must have an entropy strictly greater than $\ln K$), while we have known that among structures with K or fewer cells, the K-cell evenly partitional structure performs the best. When $\varepsilon \in (0,1)$, a $(K+1)$-cell structure is possible, leading to a structure with two cell sizes described above. ■